Rangul unei matrice

Fie $A=(a_{ij})_{\substack{1\le i\le m\\ 1\le j\le n}} \in \mathcal{M}_{m,n}(\mathbb{C})$ $(m,n\in\mathbb{N}^*)$

$$L_i \to \begin{pmatrix} \cdots & \cdots & \cdots & a_{1j} & \cdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \cdots & \cdots & \cdots & a_{mj} & \cdots \end{pmatrix}$$

$\uparrow C_j$

Să considerăm $p\in\mathbb{N}^*$, $p\le \min(m,n)$ și

$L_{i_1}, L_{i_2}, \dots, L_{i_p}$

$C_{j_1}, C_{j_2}, \dots, C_{j_p}$

cu $\{i_1, i_2, \dots, i_p\} \subseteq \{1,\dots,m\}$

și $\{j_1, j_2, \dots, j_p\} \subseteq \{1,\dots,n\}$

Def: Se numește minor de ordin p al matricei A, determinat de liniile $L_{i_1}, L_{i_2}, \dots, L_{i_p}$ și de coloanele $C_{j_1}, C_{j_2}, \dots, C_{j_p}$, determinantul matricei formate din elementele lui A situate la intersecția liniilor $L_{i_1}, \dots, L_{i_p}$ cu coloanele $C_{j_1}, \dots, C_{j_p}$.

Exemplu:

$$A = \begin{array}{c} \begin{matrix} C_1 & & C_3 & C_4 & C_5 & C_6 \end{matrix} \\ \begin{pmatrix} 1 & -1 & 0 & 1 & 2 & 3 \\ 1 & 1 & 0 & 1 & -1 & 2 \\ 1 & 1 & 9 & 4 & 5 & 3 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix} \end{array} \begin{matrix} L_1 \\ L_2 \\ L_3 \\ L_4 \end{matrix}$$

$p=2$; L_2, L_4 ; C_3, C_4

$$\begin{vmatrix} 0 & 1 \\ 0 & 1 \end{vmatrix} = 0$$

Def: Pt $A\in\mathcal{M}_{m,n}(\mathbb{C})$ și $r\in\mathbb{N}^*$, $r\le \min\{m,n\}$.

$\mathcal{O}_{m,n}$

spunem că A are rangul r ($\operatorname{rang}(A)=r$) dacă A are cel puțin un minor de ordin r nenul și orice minor de ordin $>r$ este nul.

$\operatorname{rang}(O_{m,n}) \overset{def}{=} 0$

Proprietăți ale rangului unei mat:

$${}^tA = \begin{pmatrix} 1 & 1 & 1 & 0 \\ -1 & 1 & 1 & 1 \\ 0 & 0 & 9 & 0 \\ 2 & -1 & 4 & -1 \\ 3 & 2 & 3 & 1 \end{pmatrix}$$

1° $\operatorname{rang}(A) = \operatorname{rang}({}^tA)$

2° $(\forall) A, B \in \mathcal{M}_{m,n}(\mathbb{C})$; $\operatorname{rang}(A \cdot B) \leq \begin{matrix} \operatorname{rang} A \\ \operatorname{rang} B \end{matrix}$

3° Rangul unei matrice nu se modif dacă:

transf. elementare:
- i) permutăm între ele 2 linii/coloane ale matr.
- ii) înmulțim o linie/coloană cu un număr nenul
- iii) adunăm el. unei linii/coloane la el. altei linii/coloane

Matrice inversabile

Def: Fie $A \in \mathcal{M}_n(\mathbb{C})$. Zice că A e.n. inv $(\Leftrightarrow)(\exists) B \in \mathcal{M}_n(\mathbb{C})$ a.î. $A \cdot B = B \cdot A = I_n$

În acest caz, B e.n. inv. lui A și se notează $B = A^{-1}$.

Obs: i) Inversa unei mat, dacă $(\exists)$, este unică

ii) $(A^{-1})^{-1} = A$

Teoremă: Fie $A \in \mathcal{M}_n(\mathbb{C})$. Zice A inv $(\Leftrightarrow) \det A \neq 0$

Algoritm pentru det. inversei unei matrice $A \in \mathcal{M}_n(\mathbb{C})$ cu $\det A \neq 0$

${}^tA \Rightarrow$ adjuncta lui A: $A^* = (\delta_{i,j})_{1 \leq i,j \leq n}$, unde $\delta_{ij} \overset{\text{def}}{=} (-1)^{i+j} d_{ij}$

δ_{ij}: complement algebric

d_{ij}: det. matr. obț. suprimând L_i și C_j din ${}^tA$

...me de ecuații liniare

Fie $A \in \mathcal{M}_{m,n}(\mathbb{C})$ și sist. liniar sub formă matricială

(1) $\boxed{AX = b} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \in \mathbb{C}^n$ (col. term. liberi)

$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$

coloana necunoscutelor

Teorema KRONECKER-CAPELLI.

Sist (1) este compat (=) $\operatorname{rang} A = \operatorname{rang}(\bar{A} \to A|b)$

Def: O matrice a rămas în formă scară (pe linii) dacă:

i) fiecare linie nenulă începe cu mai multe zerouri decât linia nenulă precedentă

ii) eventualele linii nule se află dedesubtul celor nenule

! Pt o matrice în formă scară (pe linii) vom numi pivot fiecare prim element nenul începând din stânga, al fiecărei linii nenule.

Exemple și contra exemple

$\left(\begin{smallmatrix} 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 \end{smallmatrix}\right)$ NU; $\left(\begin{smallmatrix} 0 & 0 & \textcircled{1} & 2 \\ 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 \end{smallmatrix}\right)$; $\left(\begin{smallmatrix} 0 & 0 & 0 & \textcircled{1} & 2 \\ 0 & 0 & 0 & 0 & \textcircled{3} \\ 0 & 0 & 0 & 0 & 0 \end{smallmatrix}\right)$; $\left(\begin{smallmatrix} \textcircled{1} \\ 3 \end{smallmatrix}\right)$; $(\textcircled{1}\ 0\ 0)$

; $\left(\begin{smallmatrix} \textcircled{2} & 2 \\ 0 & \textcircled{1} \\ 0 & 0 \end{smallmatrix}\right)$; $\left(\begin{smallmatrix} \textcircled{1} & 0 & 3 & 1 \\ 0 & 0 & 0 & \textcircled{1} \end{smallmatrix}\right)$

Ex 1: Fie sist. liniar $Ax = b$, unde $A = \begin{pmatrix} 2 & 1 & -1 & 1 & 2 \\ 4 & 2 & 2 & -1 & 1 \\ 4 & 2 & -2 & 1 & 2 \\ 7 & 3 & -1 & 1 & 5 \end{pmatrix}$

și $b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix} \in \mathbb{R}^4$

a) fol. transf. elementare pe linii, să se ~~transf~~ stabilească o condiție necesară și suficientă de compatibilitate pentru sist. liniar $Ax = b$

Obs: Dacă o mat. nenulă este în formă scară (pe linii) at. rangul matricei este egal cu nr. de linii nenule (nr. de pivoți)

a) $(A|b) = \left(\begin{smallmatrix} 2 & 1 & -1 & 1 & 2 & : & b_1 \\ 4 & 2 & 2 & -1 & 1 & : & b_2 \\ 4 & 2 & -2 & 1 & 2 & : & b_3 \\ 7 & 3 & -1 & 1 & 5 & : & b_4 \end{smallmatrix}\right) \underset{}{\overset{L_1 \leftrightarrow L_2}{\longrightarrow}}$

$$\to \left(\begin{array}{ccccc|c} 1 & 0 & 2 & -1 & 1 & b_2 \\ 2 & 1 & -1 & 1 & 2 & b_1 \\ 4 & 2 & -2 & 1 & 2 & b_3 \\ 7 & 3 & -1 & 1 & 5 & b_4 \end{array}\right) \begin{array}{l} L_2 \to L_2 + (-2)L_1 \\ \longrightarrow \\ L_3 \to L_3 + (-4)L_1 \\ L_4 \to L_4 + (-7)L_1 \end{array} \left(\begin{array}{ccccc|c} 1 & 0 & 2 & -1 & 1 & \\ 0 & \textcircled{1} & -5 & 3 & 0 & b_1 - \\ 0 & 2 & -10 & 5 & -2 & b_3 - \\ 0 & 3 & -15 & 8 & -2 & b_4 - 7b_1 \end{array}\right)$$

$$\longrightarrow \begin{array}{l} L_3 \to L_3 - 2L_2 \\ L_4 \to L_4 - 3L_2 \end{array} \left(\begin{array}{ccccc|c} \textcircled{1} & 0 & 2 & -1 & 1 & b_2 \\ 0 & \textcircled{1} & -5 & 3 & 0 & b_1 - 2b_2 \\ 0 & 0 & 0 & \textcircled{-1} & -2 & b_3 - 2b_1 \\ 0 & 0 & 0 & -1 & -2 & b_4 - b_2 - 3b_1 \end{array}\right) \longrightarrow L_4 \to L_4 - L_3$$

$$\left(\begin{array}{ccccc|c} x_1 & x_2 & x_3 & x_4 & x_5 & \\ 1 & 0 & 2 & -1 & 1 & b_2 \\ 0 & 1 & -5 & 3 & 0 & b_1 - 2b_2 \\ 0 & 0 & 0 & -1 & -2 & b_3 - 2b_1 \\ 0 & 0 & 0 & 0 & 0 & b_4 - b_2 - 3b_1 - b_3 + 2b_1 \end{array}\right) \Rightarrow$$

$$\underbrace{\left(\begin{array}{ccccc} 1 & 0 & 2 & -1 & 1 \\ 0 & 1 & -5 & 3 & 0 \\ 0 & 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)}_{U} \left|\begin{array}{c} b_2 \\ b_1 - 2b_2 \\ b_3 - 2b_1 \\ \underbrace{b_4 - b_3 - b_2 - b_1}_{\tilde{b}} \end{array}\right.$$

$\Rightarrow \operatorname{rang}(A|b) = \operatorname{rang}(U|\tilde{b})$

$\operatorname{rang} A = \operatorname{rang} U = 3$

A t rang $\bar{A} = \operatorname{rang} A \iff b_4 - b_3 - b_2 - b_1 = 0$

b) Să se rezolve sistemul $Ax = b$ pt $b = \begin{pmatrix} -1 \\ 3 \\ 0 \\ 0 \end{pmatrix}$

Nec. principale (corresp. col. cu pivot): x_1, x_2, x_4.

Nec. secundare: x_3, x_5.

$$\left(\begin{array}{ccccc|c} 1 & 0 & 2 & -1 & 1 & -1 \\ 0 & 1 & -5 & 3 & 0 & 3 \\ 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ x_1 & x_2 & x_3 & x_4 & x_5 & \end{array}\right) \begin{array}{l} \\ \\ \\ \\ \curvearrowright \end{array} \begin{array}{l} - \\ (3 \to C_4) \end{array} \left(\begin{array}{ccccc|c} 1 & 0 & -1 & 2 & 1 & -1 \\ 0 & 1 & 3 & -5 & 0 & 3 \\ 0 & 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ x_1 & x_2 & x_4 & x_3 & x_5 & \end{array}\right)$$

$$\longrightarrow \begin{array}{l} \\ L_2 \to L_2 - 3L_3 \\ L_1 \to L_1 + L_3 \end{array} \left(\begin{array}{ccccc|c} x_1 & x_2 & x_4 & x_3 & x_5 & \\ 1 & 0 & 0 & 2 & 3 & -1 \\ 0 & 1 & 0 & -5 & -6 & 3 \\ 0 & 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\overline{Ax = b} \iff \begin{cases} x_1 + 2x_3 + 3x_5 = -1 \\ x_2 - 5x_3 - 6x_5 = 3 \\ x_4 + 2x_5 = 0 \end{cases} \Leftrightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} -1 - 2x_3 - 3x_5 \\ 3 + 5x_3 + 6x_5 \\ x_3 \\ -2x_5 \\ x_5 \end{pmatrix}$$

$$\begin{pmatrix} -1 \\ 3 \\ 0 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_5 \begin{pmatrix} -3 \\ 6 \\ -4 \\ 1 \end{pmatrix}$$

combinație liniară a vectorilor $\begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}$ și $\begin{pmatrix} -3 \\ 6 \\ -4 \\ 1 \end{pmatrix}$

cu parametrii x_3 și x_5

Obs: Sol. gen. a unui sist. liniar / Nu omogen este suma între o sol. part. a acestuia și sol. gen. a sist. omogen asociat

Ex2: Aplicând o singură dată transf. elementare asupra elem. unei matrice să se rez. sim. sistemele liniare $Ax=b$ și $Ax=c$ unde $A=\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 3 & 7 & 7 \end{pmatrix}$; $b=\begin{pmatrix} 1 \\ 1 \\ 5 \end{pmatrix}$; $c=\begin{pmatrix} 2 \\ 4 \\ 4 \end{pmatrix}$

~~$A+b=\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 5 & 6 & 1 \\ 3 & 7 & 7 & 5 \end{array}\right)$~~

$$A|b|c = \left(\begin{array}{ccc|c|c} 1 & 2 & 3 & 1 & 2 \\ 2 & 5 & 6 & 1 & 4 \\ 3 & 7 & 7 & 5 & 4 \end{array}\right) \xrightarrow[L_3 \to L_3 - 3L_1]{L_2 \to L_2 - 2L_1} \left(\begin{array}{ccc|c|c} 1 & 2 & 3 & 1 & 2 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 1 & -2 & 2 & -2 \end{array}\right) \xrightarrow{L_3 \to L_3 -}$$

$$\to \left(\begin{array}{ccc|c|c} 1 & 2 & 3 & 1 & 2 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & -2 & 3 & -2 \end{array}\right) \xrightarrow{\cdot(-2)} \to \left(\begin{array}{ccc|c|c} 1 & 2 & 3 & 1 & 2 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -3/2 & 1 \end{array}\right) \xrightarrow{L_1 \to L_1 - 2L_3}$$

$$\left(\begin{array}{ccc|c|c} 1 & 2 & 0 & 11/2 & -1 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -3/2 & -2 \end{array}\right) \xrightarrow{L_1 \to L_1 - 2L_2} \left(\begin{array}{ccc|c|c} x_1 & x_2 & x_3 & & \\ 1 & 0 & 0 & 15/2 & -1 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -3/2 & -2 \end{array}\right)$$

$$Ax=b \Leftrightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 15 \\ -2 \\ -3 \end{pmatrix}; \quad Ax=c \Leftrightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Metodă de calcul pentru inversa unei matrice

Fie A o matr. pătr. de ordin n și $\det A \neq 0$ și $I_n = (e_1 | e_2 | \dots | e_n)$, unde $1 \leq k \leq n$;

$$e_k = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \leftarrow k$$

$\exists A^{-1} \overset{not}{=} (b_{ij})_{1 \le i,j \le n}$

Sistemul $Av = e_1 \Rightarrow x = A^{-1}e_1 = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ \vdots & & & \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} b_{11} \\ b_{21} \\ \vdots \\ b_{n1} \end{pmatrix}$ ← primo col a lui A^{-1}

$\Rightarrow$ pt det A^{-1} este suf. să rezolvăm sist. liniare $Ax = e_1$
$Ax = e_2$
$\dots$
$Ax = e_n$

$(A \mid I_n) \to \dots \to (I_n \mid A^{-1})$

Ex: Să se det A^{-1} fol. metoda de mai sus. $A = \begin{pmatrix} 2 & 1 & -1 \\ 3 & 1 & 1 \\ 5 & 2 & 1 \end{pmatrix}$

$$(A \mid I_3) = \left(\begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 3 & 1 & 1 & 0 & 1 & 0 \\ 5 & 2 & 1 & 0 & 0 & 1 \end{array}\right) \overset{L_2 \to L_2 - L_1}{\to}$$

$$\to \left(\begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 1 & 0 & 2 & -1 & 1 & 0 \\ 5 & 2 & 1 & 0 & 0 & 1 \end{array}\right) \overset{L_1 \leftrightarrow L_2}{\to} \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & -1 & 1 & 0 \\ 2 & 1 & -1 & 1 & 0 & 0 \\ 5 & 2 & 1 & 0 & 0 & 1 \end{array}\right) \overset{\substack{L_2 \to L_2 - 2L_1 \\ L_3 \to L_3 - 5L_1}}{\to}$$

$$\to \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & -1 & 1 & 0 \\ 0 & 1 & -5 & 3 & -2 & 0 \\ 0 & 2 & -9 & 5 & -5 & 1 \end{array}\right) \to \overset{L_3 \to L_3 - 2L_2}{\left(\begin{array}{ccc|ccc} 1 & 0 & 2 & -1 & 1 & 0 \\ 0 & 1 & -5 & 3 & -2 & 0 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array}\right)}$$

$$\to \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 3 & 2 \\ 0 & 1 & 0 & -2 & -7 & 5 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array}\right)$$

A^{-1}

Temă: $A^{-1} = ?$ pt $A = \begin{pmatrix} 3 & 1 & -2 \\ -7 & -2 & 3 \\ 4 & 1 & 1 \end{pmatrix}$

Rezolvare: $A \mid I_3 = \left(\begin{array}{ccc|ccc} 3 & 1 & -2 & 1 & 0 & 0 \\ -7 & -2 & 3 & 0 & 1 & 0 \\ 4 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \overset{L_3 \to L_3 - L_1}{\to}$

$$\to \left(\begin{array}{ccc|ccc} 3 & 1 & -2 & 1 & 0 & 0 \\ -7 & -2 & 3 & 0 & 1 & 0 \\ 1 & 0 & 3 & -1 & 0 & 1 \end{array}\right) \overset{L_1 \leftrightarrow L_3}{\to} \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & -1 & 0 & 1 \\ -7 & -2 & 3 & 0 & 1 & 0 \\ 3 & 1 & -2 & 1 & 0 & 0 \end{array}\right) \overset{\substack{L_2 \to L_2 + 2L_1 \\ L_3 \to L_3 - 3L_1}}{\to}$$

$$\to \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & -1 & 0 & 1 \\ 0 & -2 & 24 & -7 & 1 & 7 \\ 0 & 1 & -11 & 4 & 0 & -3 \end{array}\right) \overset{L_2 \leftrightarrow L_3}{\to} \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & -1 & 0 & 1 \\ 0 & 1 & -11 & 4 & 0 & -3 \\ 0 & -2 & 24 & -7 & 1 & 7 \end{array}\right) \overset{L_3 \to L_3 + 2L_2}{\to}$$

12.10.2021

ALGA ED-S2

1) Fie $V = \{ P \in \mathbb{R}_3[X] \mid P(-1) + P'(1) = 0 \} \subseteq \mathbb{R}_3[X]$

a) Să se arate că $V \leq \mathbb{R}_3[X]$

b) Să se determine o bază pt V

c) Să se determine coordonatele polinomului $\hat{R}=$ în baza de la b)

Sol: a) Fie $\alpha, \beta \in \mathbb{R}$ (pt că $\mathbb{R}_3[X]$ este sp. vect. real)

și $P, Q \in V$. Vrem să arătăm că $\alpha P + \beta Q \in V$

$(\alpha P+\beta Q)(-1)+(\alpha P+\beta Q)'(1) = \alpha \underbrace{P(-1)}+\beta Q(-1) + \alpha \underbrace{P'(1)}+$
$\beta Q'(1)$

$= \alpha \underbrace{(P(-1) + P'(1)}_{0} + \beta \underbrace{(Q(-1)+Q'(1))}_{=0} = 0 \Rightarrow \alpha P+\beta Q \in V$

Deci, $V \leq \mathbb{R}_3[X]$

b) Fie $P = a_0 + a_1 X + a_2 X^2 + a_3 X^3 \in \mathbb{R}_3[X]$

$P' = a_1 + 2a_2 X + 3a_3 X^2$

$P \in V \Leftrightarrow P(-1)+P'(1)=0 \Leftrightarrow a_0 - a_1 + a_2 - a_3 + a_1 + 2a_2 + 3a_3 = 0$

$\Leftrightarrow \boxed{a_0 + 3a_2 + 2a_3 = 0}$

$\Leftrightarrow a_0 = -3a_2 - 2a_3$

$\Downarrow$

$V = \{ \underbrace{-3a_2 - 2a_3 + a_1 X + a_2 X^2 + a_3 X^3}_{} \mid a_1, a_2, a_3 \in \mathbb{R} \}$

$a_1 \underbrace{X}_{P_1} + a_2 \underbrace{(-3+X^2)}_{P_2} + a_3 \underbrace{(-2+X^3)}_{P_3}$

$\Rightarrow \{P_1, P_2, P_3\}$ SG pt V.

Obs: (V) mulțime de vectori polinoame de grade 2 câte 2 diferite este sist. lin. $\Leftrightarrow$ independent.

$\Rightarrow \{P_1, P_2, P_3\}$ bază pt V și $\dim_{\mathbb{R}} V = 3$

c) $R = \underbrace{x^3 - 3x^2 + 2x + 7}_{''}$

$1 \cdot (x^2+2) + (-3)\cdot(x^2-3) + 2x \Rightarrow (R)_B = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} \begin{pmatrix} P_3 \\ P_2 \\ P_1 \end{pmatrix} \Rightarrow \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} \begin{pmatrix} P_1 \\ P_2 \\ P_3 \end{pmatrix}$

Obs: $x^3 - 3x^2 + 2x + 7 = a_1 X + a_2(-3+x^2) + a_3(-2+x^3) \Rightarrow \ldots$ (se identif. coef.)

Exercițiul 2:

Fie $B_1 = \{A = 1 + 2x + x^2, B = -1 + x^2, C = 1 + x + 2x^2\} \subseteq \mathbb{R}_2[X]$
și $B_2 = \{D = 1 + x^2, E = 2 + x + x^2, F = 1 - x + x^2\} \subseteq \mathbb{R}_2[X]$.

a) Să se arate că B_1 și B_2 sunt baze pentru $\mathbb{R}_2[X]$
b) Știind că pt polin $Q \in \mathbb{R}_2[X]$ avem coord lui a în baza B_1 sunt $\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$, să se det coord lui Q în baza B_2.

a) $\dim \mathbb{R}_2[X] = 3$
B_1, B_2 conț câte 3 vectori $\Big| \xRightarrow{\text{curs}}$ pt a dem că B_1, B_2 sunt bază ale lui $\mathbb{R}_2[X]$ este suficient să dem că fiecare este SLI.

B_1 SLI $\xRightarrow{\text{curs}}$ matr. coordonatelor vectorilor din B_1 în baza canonică ($B_c = \{1, X, X^2\}$) a lui $\mathbb{R}_2[X]$ este inversabilă $\Rightarrow \det \neq 0$.

$\det A_1 = \begin{vmatrix} 1 & -1 & 1 \\ 2 & 0 & 1 \\ 1 & 1 & 2 \end{vmatrix} = 0 + 2 - 1 - 1 + 4 = 4 \neq 0 \Rightarrow B_1$ SLI

A_1 / matr. coresp lui B_1

$\Rightarrow B_1$ bază pt $\mathbb{R}_3[X]$

Asemănător pt B_2:

$\det A_2 = \begin{vmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{vmatrix} = 1 + 0 - 2 - 1 + 1 = -1 \neq 0$

$\Rightarrow B_2$ SLI $\Rightarrow B_2$ bază pt $\mathbb{R}_2[X]$

b) $(Q)_{B_1} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \Rightarrow Q = 3A + 2B + C$

$= 3(1 + 2x + x^2) + 2(-1 + x^2) + 1 + x + 2x^2$

$= 2 + 7x + 7x^2$

$Q = \alpha D + \beta E + \gamma F \Rightarrow 2 + 7x + 7x^2 = \alpha(1 + x^2) + \beta(2 + x + x^2) + \gamma(1 - x + x^2)$

$\Rightarrow 2 + 7x + 7x^2 = (\alpha + 2\beta + \gamma) + (\beta - \gamma)x + (\alpha + \beta + \gamma)x^2$

$$\begin{cases} 2 = \alpha + 2\beta + \gamma \\ 7 = \beta - \gamma \\ 7 = \alpha + \beta + \gamma \end{cases} \quad \text{"-"} \Rightarrow \beta = -5$$

$$\Rightarrow -\gamma = 7 + 5 \Rightarrow \gamma = -12 \Rightarrow \alpha = 24$$

$$2)\ (a)_{\mathcal{B}_2} = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 24 \\ -5 \\ -12 \end{pmatrix}$$

Ex 3. Să se determine o bază și dim pt $V = \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_5 \end{pmatrix} \in \mathbb{R}^5 \;\middle|\; \begin{cases} x_1 + 2x_2 + 3x_3 - x_4 + x_5 = 0 \\ -2x_1 + 3x_2 + 7x_3 + 2x_4 - x_5 = 0 \\ x_1 + 9x_2 + 16x_3 - x_4 + 2x_5 = 0 \\ x_1 + 23x_2 + 42x_3 - x_4 + 4x_5 = 0 \end{cases} \right\}$

(=) aducem la forma scară

$$A = \begin{pmatrix} 1 & 2 & 3 & -1 & 1 \\ -2 & 3 & 7 & 2 & -1 \\ 1 & 9 & 16 & -1 & 2 \\ 1 & 23 & 42 & -1 & 4 \end{pmatrix} \begin{matrix} \\ L_2 \to L_2 + 2L_1 \\ L_3 \to L_3 - L_1 \\ L_4 \to L_4 - L_1 \end{matrix}$$

$(\dim V = \dim \ker(A))$

$$\to \begin{pmatrix} 1 & 2 & 3 & -1 & 1 \\ 0 & 7 & 13 & 0 & 1 \\ 0 & 7 & 13 & 0 & 1 \\ 0 & 21 & 39 & 0 & 3 \end{pmatrix} \begin{matrix} \\ \\ L_3 \to L_3 - L_2 \\ L_4 \to L_4 - 3L_2 \end{matrix}$$

$$\to \begin{pmatrix} \boxed{1} & 2 & 3 & -1 & 1 \\ 0 & \boxed{7} & 13 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1, x_2 \text{ nec. p.} \\ x_3, x_4, x_5 \text{ parametri} \end{pmatrix}$$

$$\begin{matrix} x_1 & x_3 & x_3 & x_4 & x_2 \end{matrix}$$

$$C_2 \leftrightarrow C_5 \to \begin{pmatrix} \boxed{1} & 3 & 3 & -1 & 2 \\ 0 & \boxed{1} & 13 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{matrix} x_1, x_5 \text{ nec. p.} \\ x_3, x_4, x_2 \text{ parametri} \end{matrix}$$

$$\begin{matrix} x_1 & x_5 & x_3 & x_4 & x_2 \end{matrix}$$

$$L_1 \to L_1 - L_2 \to \begin{pmatrix} \boxed{1} & 0 & -10 & -1 & -5 \\ 0 & \boxed{1} & 13 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{matrix} x_1 = 5x_2 + 10x_3 + x_4 \\ x_5 = -7x_2 - 13x_3 \end{matrix}$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 5x_2 + 10x_3 + x_4 \\ x_2 \\ x_3 \\ x_4 \\ -7x_2 - 13x_3 \end{pmatrix} =$$

$$= x_2 \underbrace{\begin{pmatrix} 5 \\ 1 \\ 0 \\ 0 \\ -7 \end{pmatrix}}_{v_1} + x_3 \underbrace{\begin{pmatrix} 10 \\ 0 \\ 1 \\ 0 \\ -13 \end{pmatrix}}_{v_2} + x_4 \underbrace{\begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}}_{v_3}$$

$\Rightarrow \{v_1, v_2, v_3\}$ bază pt V

$\dim_{\mathbb{R}} V = 3$

V pt $\mathbb{R}^5$ $\Rightarrow$ $\{e_1, e_2, e_3, e_4, e_5\}$ bază canonică $\Rightarrow$ $\{e_1, e_2, e_3, e_4\}$ liniar independente

Ex 4: Fie V un $\mathbb{K}$-sp. vect și $V_1, V_2 \subset V$.

Definim $V_1 + V_2 \overset{\text{def}}{=} \{ v_1 + v_2 \mid v_1 \in V_1, v_2 \in V_2 \}$

$V_1 \cap V_2 = \{ v \mid v \in V_1, v \in V_2 \}$

Să se arate că $V_1 + V_2$ și $V_1 \cap V_2$ sunt subspații vectoriale.

↑ PENTRU EXAMEN

Fie $\alpha, \beta \in \mathbb{K}$, $v_1 + v_2 \in V_1 + V_2$ ($v_1 \in V_1$, $v_2 \in V_2$), $w_1 + w_2 \in V_1 + V_2$ ($w_1 \in V_1$, $w_2 \in V_2$)

A.: $\alpha(v_1 + v_2) + \beta(w_1 + w_2) = \underbrace{\alpha v_1 + \beta w_1}_{\in V_1} + \underbrace{\alpha v_2 + \beta w_2}_{\in V_2} \in V_1 + V_2$

$\Rightarrow V_1 + V_2$ subsp V.

$V_1 + V_2 \subset V$

Fie $\alpha, \beta \in \mathbb{K}$ și $u, v \in V_1 \cap V_2 \Rightarrow u, v \in V_1 \overset{V_1 \subset V}{\Rightarrow} \alpha u + \beta v \in V_1$

A.: $\alpha u + \beta v$ $\Rightarrow u, v \in V_2 \overset{V_2 \subset V}{\Rightarrow} \alpha u + \beta v \in V_2$

$\Rightarrow \alpha u + \beta v \in V_1 \cap V_2 \Rightarrow$

Exs:

Să se stabilească pentru fiecare din urm. mulțimi dacă este subsp. vect în sp vect indicat:

a) $V_1 = \{ (x_1, x_2) \in \mathbb{R}^2 \mid x_1 \cdot x_2 = 0 \} \subset \mathbb{R}^2$ ✗

b) $V_2 = \{ P \in \mathbb{R}_2[X] \mid P(1) = 2 \}$ subm. $\subseteq \mathbb{R}_2[X]$ ✗

c) $V_3 = \{ A \in \mathcal{M}_2(\mathbb{R}) \mid A = {}^tA \}$ ✓

d) $V_4 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid \begin{matrix} 2x + y = 2 \\ y - z = 1 \end{matrix} \right\}$ ✗

obs: $0_V \in V_1$

Fie $u \in V_1$ at $\underset{\in V_1}{\underbrace{0_{\mathbb{K}} \cdot u}} = 0_V$

---

a) $(1,0), (0,1) \in V_1$, dar $(1,0) + (0,1) = (1,1) \notin V_1$

$\Rightarrow V_1 \not\subset \mathbb{R}^2$

b) Dacă $V_2 \subset \mathbb{R}_2[X]$ ș) $0_{\mathbb{R}_2[X]} \in V_2$, dar $0_{\mathbb{R}_2[X]}(1) = 0 \neq 2$

$\Rightarrow V_2 \not\subset \mathbb{R}_2[X]$

c) $V_3 = \{A \in \mathcal{M}_2(\mathbb{R}) \mid A = {}^tA\} \subseteq \mathcal{M}_2(\mathbb{R})$

Fie $\alpha, \beta \in \mathbb{R}$ și $A, B \in V_3$ ⇒ $A = {}^tA$, $B = {}^tB$.

At. ${}^t(\alpha A + \beta B) = {}^t(\alpha A) + {}^t(\beta B)$

$= \alpha\, {}^tA + \beta\, {}^tB = \alpha A + \beta B$

$\Rightarrow \alpha A + \beta B \in V_3 \Rightarrow V_3 \subseteq \mathcal{M}_2(\mathbb{R})$

d) $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \notin V_4 \Rightarrow V_4 \not\subseteq \mathbb{R}^3$

Ex6: Fie V un $\mathbb{K}$-sp.ved și $V_1, V_2 \subseteq V$. Atunci:

$V_1 \cup V_2 \subseteq V \Leftrightarrow V_1 \subseteq V_2$ sau $V_2 \subseteq V_1$

Dem

"$\Leftarrow$" Evident

Dacă $V_1 \subseteq V_2 \Rightarrow V_1 \cup V_2 = V_2 \subseteq V$

$V_2 \subseteq V_1 \Rightarrow V_1 \cup V_2 = V_1 \subseteq V$

"$\Rightarrow$" Presupunem $V_1 \not\subseteq V_2$ și $V_2 \not\subseteq V_1$. Atunci $\exists\, x \in V_1, x \notin V_2 \Rightarrow x \in V_1 \cup V_2$

$y \in V_2, y \notin V_1 \Rightarrow y \in V_1 \cup V_2$

~~$x + y \in V_1 \cup V_2 \subseteq V$~~

$V_1 \cup V_2 \subseteq V$

$\Rightarrow x + y \in V_1 \cup V_2$

Prin RA să:

- $x + y \in V_1$, dar $x \in V_1 \Rightarrow -x \in V_1$, $V_1 \subseteq V \Rightarrow (x+y) - x \in V_1 \Rightarrow y \in V_1$ ↯ cu $y \notin V_1$
- $x + y \in V_2$, dar $y \in V_2 \Rightarrow -y \in V_2$, $V_2 \subseteq V \Rightarrow (x+y) - y \in V_2 \Rightarrow x \in V_2$ ↯ cu $x \notin V_2$

ALGAED-S3

1) Fie V sp { $P_1 = 1+2x+3x^2$, $P_2 = e+3x+4x^2$, $P_3 = 2+x$, $P_4 = 3+2x+3x^2$, $P_5 = 4+3x+4x^2$ } $\subset \mathbb{R}_2[x]$.

Să se det dim și o bază.

$$A = \begin{pmatrix} P_1 & P_2 & P_3 & P_4 & P_5 \\ 1 & 2 & 2 & 3 & 4 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \end{pmatrix} \xrightarrow[L_3 \to L_3 - 3L_1]{L_2 \to L_2 - 2L_1} \begin{pmatrix} 1 & 2 & 2 & 3 & 4 \\ 0 & -1 & -3 & -4 & -5 \\ 0 & 2 & -2 & 0 & 0 \end{pmatrix} \xrightarrow{L_3 \to L_3 + 2L_2} \begin{pmatrix} 1 & 2 & 2 & 3 & 4 \\ 0 & -1 & -3 & -4 & -5 \\ 0 & 0 & -8 & -8 & -10 \end{pmatrix}$$

$$\longrightarrow \begin{pmatrix} P_1 & P_2 & P_3 & P_4 & P_5 \\ 1 & 2 & 2 & 3 & 4 \\ 0 & -1 & -3 & -4 & -5 \\ 0 & 0 & 2 & 2 & 2 \end{pmatrix} \xrightarrow{P_3 \leftrightarrow P_4} \begin{pmatrix} P_1 & P_2 & P_4 & P_3 & P_5 \\ 1 & 2 & 3 & 2 & 4 \\ 0 & -1 & -4 & -3 & -5 \\ 0 & 0 & 2 & 2 & 2 \end{pmatrix}$$

$$\xrightarrow{L_3 \to L_3/2} \begin{pmatrix} 1 & 2 & 3 & 2 & 4 \\ 0 & -1 & -4 & -3 & -5 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix} \xrightarrow[L_1 \to L_1 - 3L_3]{L_2 \to L_2 + 4L_3} \begin{pmatrix} 1 & 2 & 0 & -1 & 1 \\ 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$

$$\xrightarrow{L_1 \to L_1 - 2L_2} \left(\begin{array}{ccc|cc} P_1 & P_2 & P_4 & P_3 & P_5 \\ 1 & 0 & 0 & 1 & -1 \\ 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{array}\right)$$

prescurtări de calcul.

$$\xrightarrow{\cdot} \begin{pmatrix} 1 & 2 & 2 & 3 & 4 \\ 0 & -1 & -3 & -4 & -5 \\ 0 & 0 & 0 & 2 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 3 & 2 & 4 \\ 0 & 1 & 4 & 3 & 5 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

$$\longrightarrow \begin{pmatrix} 1 & 2 & 0 & 2 & 1 \\ 0 & 1 & 0 & 3 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix} \longrightarrow \left(\begin{array}{ccc|cc} P_1 & P_2 & P_4 & P_3 & P_5 \\ 1 & 0 & 0 & -4 & -1 \\ 0 & 1 & 0 & 3 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{array}\right)$$

2) $\{P_1, P_2, P_4\}$ o bază pentru V

B bază

$\dim_{\mathbb{R}} V = 3$

$P_3 = -4P_1 + 3P_2 \Rightarrow (P_3)_B = \begin{pmatrix} -4 \\ 3 \\ 0 \end{pmatrix}$ coloana coord lui P_3 în baza B

$P_5 = -P_1 + P_2 + P_4 \Rightarrow (P_5)_B = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$

21 U, V ~~a probe~~ $\subset \mathbb{R}^4$

$U = sp\{u_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 5 \end{pmatrix}, u_2 = \begin{pmatrix} 1 \\ 1 \\ 2 \\ 2 \end{pmatrix}, u_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}\} \subset \mathbb{R}^4$

și $V = sp\{v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ a \end{pmatrix}, v_2 = \begin{pmatrix} 2 \\ 1 \\ 2 \\ 2 \end{pmatrix}\} \subset \mathbb{R}^4$

Arătați că $U \oplus V = \mathbb{R}^4$. Pentru $w = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix}$ să se det u în U și v în V a.î. $w = u+v$.

Obs ~~Dacă $v_1, v_2 \in$~~

Dacă $U \leq V$ GW (K-sp vect)

și $U = \mathrm{sp}\{u_1, \dots, u_p\}$, iar $V = \mathrm{sp}\{v_1, \dots, v_q\}$ atunci

$U + V = \mathrm{sp}\{u_1, \dots, u_p, v_1, \dots, p_q\}$

$$A = \begin{matrix} u_2 & u_3 & u_1 & v_1 & v_2 \end{matrix} \begin{pmatrix} 1 & 0 & 2 & 1 & 5 \\ 1 & 1 & 3 & 1 & 2 \\ 5 & 1 & 11 & 3 & 6 \\ 2 & 1 & 5 & 4 & 2 \end{pmatrix} \xrightarrow[L_4 \to L_4 - 2L_1]{L_2 \to L_2 - L_1,\ L_3 \to L_3 - 5L_1} \begin{matrix} u_2 & u_3 & u_1 & v_1 & v_2 \end{matrix} \begin{pmatrix} 0 & 0 & 2 & 1 & 5 \\ 0 & 0 & 1 & 0 & -3 \\ 0 & 1 & 1 & -2 & -19 \\ 0 & 1 & 1 & 2 & -8 \end{pmatrix}$$

$$\xrightarrow{L_3 \to L_3 - L_2,\ L_4 \to L_4 - L_2} \begin{pmatrix} 1 & 0 & 2 & 1 & 5 \\ 0 & 1 & 1 & 0 & -3 \\ 0 & 0 & 0 & -2 & -16 \\ 0 & 0 & 0 & 2 & -5 \end{pmatrix} \xrightarrow{C_4 \leftrightarrow C_3} \begin{matrix} u_2 & u_3 & v_1 & u_1 & v_2 \end{matrix} \begin{pmatrix} 1 & 0 & 0 & 2 & 5 \\ 0 & 1 & 0 & 1 & -3 \\ 0 & 0 & +1 & 0 & +8 \\ 0 & 0 & 2 & 0 & -5 \end{pmatrix}$$

$$\xrightarrow{L_4 \to L_4 - 2L_3} \begin{matrix} u_2 & u_3 & v_1 & u_1 & v_2 \end{matrix} \begin{pmatrix} 1 & 0 & 1 & 2 & 5 \\ 0 & 1 & 0 & 1 & -3 \\ 0 & 0 & 1 & 0 & 8 \\ 0 & 0 & 0 & 0 & -21 \end{pmatrix} \xrightarrow{:(-21)} \begin{pmatrix} 1 & 0 & 1 & 2 & 5 \\ 0 & 1 & 0 & 1 & -3 \\ 0 & 0 & 1 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow{C_4 \leftrightarrow C_5} \begin{matrix} u_2 & u_3 & v_1 & v_2 & u_1 \end{matrix} \begin{pmatrix} 1 & 0 & 1 & 5 & 2 \\ 0 & 1 & 0 & -3 & 1 \\ 0 & 0 & 1 & 8 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

$\Rightarrow$ o bază pt $U+V$ este $\{u_2, u_3, v_1, v_2\}$

$\Rightarrow \dim_{\mathbb{R}}(U+V) = 4$

$\Rightarrow \{v_1, v_2\}$ bază pt V și $\dim_{\mathbb{R}} V = 2$

dar $U+V \leq \mathbb{R}^4$ și $\dim_{\mathbb{R}} \mathbb{R}^4 = 4$ $\Rightarrow$

$\Rightarrow U+V = \mathbb{R}^4$

$u_1 = 2u_2 + u_3$ | $\{u_1, u_2, u_3\}$ SG pt U

$\{u_2, u_3\}$ SLI | $\Rightarrow$ $\{u_1, u_2\}$ bază pt U și $\dim_{\mathbb{R}} U = 2$

$\dim_{\mathbb{R}} U = \dim_{\mathbb{R}} V = 2$, $\dim_{\mathbb{R}}(U+V) = 4$ și $U+V = \mathbb{R}^4$ $\xrightarrow{\text{t. GRASSMANN}}$

$\dim_{\mathbb{R}}(U \cap V) = 0 \Leftrightarrow U \cap V = \{0_{\mathbb{R}^4}\} \Rightarrow U+V = U \oplus V$

$\Rightarrow U \oplus V = \mathbb{R}^4$.

$w = \underbrace{a u_2 + b u_3}_{u} + \underbrace{c v_1 + d v_2}_{v}$

$$(*)\begin{pmatrix} 2 \\ 3 \\ 0 \\ 2 \end{pmatrix} = a\begin{pmatrix} 1 \\ 1 \\ 5 \\ 2 \end{pmatrix} + b\begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix} + c\begin{pmatrix} 1 \\ 1 \\ 3 \\ 4 \end{pmatrix} + d\begin{pmatrix} 5 \\ 2 \\ 6 \\ 2 \end{pmatrix}$$

$$(*)\begin{cases} 2 = a + c + 5d \\ 0 = a + b + c + 2d \\ 0 = 5a + b + 3c + 6d \\ 3 = 2a + b + 4c + 2d \end{cases} \xrightarrow{④-②} a + 3c = 3 \Rightarrow -a = 3c - 3$$

$④ - ③ \quad -3a + c - 4d = 3$

$\Rightarrow 3(3c-3) + c - 4d = 3$

$\Rightarrow 10c - 4d = 12 \Rightarrow 6 = 5c - 2d.$

$b = -a - c - 2d$

$= 3c - 3c + 6 - 5c$

$\Rightarrow \boxed{b = -2c + 3}$

În ①: $2 = 3 - 2c + \frac{5}{2}(5c - 6)$

$\Rightarrow c = \frac{4}{3} \Rightarrow d = \frac{1}{3} \Rightarrow a = b = -1$

$\Rightarrow w = \underbrace{(-1)u_2 + (-1)u_3}_{u} + \underbrace{\frac{4}{3}v_1 + \frac{1}{3}v_2}_{v}$

$v = \frac{1}{3}\begin{pmatrix} 9 \\ 6 \\ 12 \\ 6 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 4 \\ 2 \end{pmatrix}$

$\begin{pmatrix} -1 \\ -2 \\ -6 \\ -3 \end{pmatrix}$

3) Fie V un $\mathbb{K}$-sp vectorial cu $\dim_{\mathbb{K}} V = n \ (\geq 2)$ și $V_1, V_2 \leq V$ a.î. $V_1 \neq V_2$ și $\dim_{\mathbb{K}} V_1 = \dim_{\mathbb{K}} V_2 = n-1$. Să se determine $\dim_{\mathbb{K}}(V_1 \cap V_2)$

$V_1, V_2 \leq V \Rightarrow V_1, V_2 \leq V_1 + V_2 \leq V$

$\dim_{\mathbb{K}}(V_1 + V_2) \in \{n-1, n\}$

Dacă $\dim_{\mathbb{K}}(V_1 + V_2) = n-1$

cum $V_1, V_2 \leq V_1 + V_2$ și $\dim_{\mathbb{K}} V_1 = \dim_{\mathbb{K}} V_2 = n-1$ $\Rightarrow V_1 = V_1 + V_2$, $V_2 = V_1 + V_2$ $\Rightarrow$ con

2) Só $\dim_{\mathbb{K}}(V_1+V_2) \neq n-1$ (mas fá-lo-ia este falso) $\Leftrightarrow$

$\Leftrightarrow \dim_{\mathbb{K}}(V_1+V_2) = n$

2) $\dim_{\mathbb{K}}(V_1 \cap V_2) = 2n-2-n = \underline{n-2}$

4) Em $\mathcal{M}_2(\mathbb{R})$ as cons. caracter. ved:

$V_1 = Sp\left\{A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}, B = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}\right\} \leq \mathcal{M}_2(\mathbb{R})$ 3i

$V_2 = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathcal{M}_2(\mathbb{R}) \;\middle|\; \begin{matrix} a+b+2d=0 \\ 2a-c+d=0 \end{matrix} \right\} \leq \mathcal{M}_2(\mathbb{R})$

Só se det. cõe o basõ 3i dim pt $V_1, V_2, V_1+V_2, V_1 \cap V_2$

Obs cõ $\{A,B\}$ este a basõ pt V_1 ($\Rightarrow \dim_{\mathbb{R}} V_1 = 2$.

$$\begin{cases} a+b+2d=0 \\ 2a-c+d=0 \end{cases} (=)$$

$$\begin{matrix} a & b & c & d \end{matrix}$$
$$\begin{pmatrix} 1 & 1 & 0 & 2 \\ 2 & 0 & -1 & 1 \end{pmatrix} \xrightarrow{L_2 \to L_2 - 2L_1} \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & -2 & -1 & -3 \end{pmatrix} \xrightarrow{b \leftrightarrow c} \underbrace{\begin{pmatrix} a & c & b & d \\ 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 3 \end{pmatrix}}_{\text{mat principal}} \cdot(-1)$$

$$(=) \begin{cases} a = -b-2d \\ c = -2b-3d \end{cases}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -b-2d & b \\ -2b-3d & d \end{pmatrix} = b \underbrace{\begin{pmatrix} -1 & 1 \\ -2 & 0 \end{pmatrix}}_{C} + d \underbrace{\begin{pmatrix} -2 & 0 \\ -3 & 1 \end{pmatrix}}_{D}$$

2) $\{C,D\}$ o basõ pt V_2 3i' $\dim_{\mathbb{R}} V_2 = 2$

$$\begin{pmatrix} 1 & 2 & -1 & -2 \\ -1 & 1 & 1 & 0 \\ 1 & -2 & 0 & -3 \\ 0 & 0 & 2 & 1 \end{pmatrix} \xrightarrow[L_4 \to L_4 - 3L_1]{\substack{L_2 \to L_2 + L_1 \\ L_3 \to L_3 - 2L_1}} \begin{pmatrix} 1 & 2 & -1 & -2 \\ 0 & 3 & 0 & -2 \\ 0 & -3 & 0 & 1 \\ 0 & -6 & 3 & 7 \end{pmatrix} \xrightarrow[L_4 \to L_4 + 2L_2]{L_3 \to L_3 + L_2}$$

$$\to \begin{pmatrix} \textcircled{1} & 2 & -1 & -2 \\ 0 & \textcircled{3} & 0 & -2 \\ 0 & 0 & \textcircled{3} & -1 \\ 0 & 0 & 3 & 3 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & -1 & -2 \\ 0 & 3 & 0 & -2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & -1 \end{pmatrix} \to \begin{pmatrix} \textcircled{1} & 2 & -1 & -2 \\ 0 & \textcircled{2} & 0 & -2 \\ 0 & 0 & \textcircled{1} & -1 \\ 0 & 0 & 0 & \textcircled{1} \end{pmatrix}$$
$$A \quad B \quad C \quad D$$

$\Rightarrow \{A,B,C,D\}$ o basõ pt $V_1+V_2 \Rightarrow$

$\Rightarrow \dim_{\mathbb{R}}(V_1+V_2) = 4 = \dim_{\mathbb{R}} \mathcal{M}_2(\mathbb{R})$ 2) $V_1+V_2 = \mathcal{M}_2(\mathbb{R})$

$\Rightarrow \dim_{\mathbb{R}}(V_1 \cap V_2) = 0$

$\Rightarrow V_1 \cap V_2 = \left\{ \begin{pmatrix} 0 & v \\ 0 & 0 \end{pmatrix} \right\}$ și $\underline{V_1 \oplus V_2 = \mathcal{M}_2(\mathbb{R})}$

5) Fie $V_1 = \mathcal{L}\{P = 1+x+x^3, Q = 2-x^2, R = x-x^3\} \subset \mathbb{R}_3[X]$
și $V_2 = \{P \in \mathbb{R}_3[X] \mid P(-1) + P'(1) = 0\} \subset \mathbb{R}_3[X]$.
Să se afle câte o bază și dim pt $V_1, V_2, V_1+V_2, V_1 \cap V_2$.

Fie $P = a_0 + a_1 X + a_2 X^2 + a_3 X^3$

$\Rightarrow P' = a_1 + 2a_2 X + 3a_3 X^2$

$P \in V_2 \Rightarrow 0 = P(-1) + P'(1)$

$= a_0 - a_1 + a_2 - a_3 + a_1 + 2a_2 + 3a_3$

$\Rightarrow 0 = a_0 + 3a_2 + 2a_3 \Rightarrow a_0 = -3a_2 - 2a_3$

$\Rightarrow V_2 = \{-3a_2 - 2a_3 + a_1 X + a_2 X^2 + a_3 X^3 \mid a_1, a_2, a_3 \in \mathbb{R}\}$

$a_1(X) + a_2(-3+X^2) + a_3(-2+X^3)$

$\begin{cases} S = X \\ T = -3 + X^2 \\ U = -2 + X^3 \end{cases}$ o bază pt V_2

$$A = \begin{pmatrix} 1 & 2 & 0 & 0 & -3 & -2 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 & 0 & 1 \\ P & Q & R & S & T & U \end{pmatrix} \begin{matrix} \\ L_2 \to L_2 - L_1 \\ \\ L_4 \to L_4 - L_1 \\ \\ \end{matrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & 0 & -3 & -2 \\ 0 & -2 & 1 & 1 & 3 & 2 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & -2 & -1 & 0 & 3 & 3 \end{pmatrix} \xrightarrow{L_3 \leftrightarrow L_2} \begin{pmatrix} 1 & 2 & 0 & 0 & -3 & -2 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & -2 & 1 & 1 & 3 & 2 \\ 0 & -2 & -1 & 0 & 3 & 3 \end{pmatrix} \begin{matrix} \\ L_3 \to L_3 - 2L_2 \\ \\ L_4 \to L_4 - 2L_2 \end{matrix}$$

$$\to \begin{pmatrix} 1 & 2 & 0 & 0 & -3 & -2 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & -1 & 0 & 1 & 3 \end{pmatrix} \xrightarrow{L_4 \to L_4 + L_3} \begin{pmatrix} 1 & 2 & 0 & 0 & -3 & -2 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 & 5 \end{pmatrix} \begin{matrix} L_1 \to L_1 - 2L_2 \\ \\ \\ L_3 \to L_3 - L_4 \end{matrix}$$

$$\to \begin{pmatrix} 1 & 0 & 0 & 0 & -1 & -2 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & -3 \\ 0 & 0 & 0 & 1 & 2 & 5 \\ P & Q & R & S & T & U \end{pmatrix}$$

2) $\{P, Q, R\}$ bază pt $V_1 + V_2$ și $V_1 + V_2 = \mathbb{R}_3[X]$
(și au aceeași dim, deci coincid)

$\{P, Q, R\}$ bază pt V_1

TG $\Rightarrow \dim_{\mathbb{R}}(V_1 \cap V_2) = 3 + 3 - 4 = 2$

$T = -P - Q - R + 2S$ (a) $\underbrace{2S - T}_{\in V_2} = \underbrace{P + Q + R}_{\in V_1} \in V_1 \cap V_2$

$U = 2P - 3R + 5S$ (b) $\underbrace{2P + 3R}_{\in V_2} = \underbrace{5S - U}_{\in V_1} \in V_1 \cap V_2$

$P + Q + R = 3 + X - X^2$

$2P + 3R = 2(1 + X + X^3) + 3(X - X^3) = 2 + 5X - X^3$

$\Rightarrow \{3 + X - X^2, 2 + 5X - X^3\} \in V_1 \cap V_2$

SLi

$\dim_{\mathbb{R}}(V_1 \cap V_2) = 2$

$\Rightarrow \{3 + X - X^2, 2 + 5X - X^3\}$ baza pt $V_1 \cap V_2$.

ALG AED - §4

Ex 1: Fie $\vec{a}, \vec{b}, \vec{c} \in V_3$. Atunci

$\{\vec{a}, \vec{b}, \vec{c}\}$ SLi $\Leftrightarrow$ $\{\vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}\}$ SLi

Sol

„$\Rightarrow$" Fie $\alpha, \beta, \gamma \in \mathbb{R}$ a.î. $\alpha(\vec{a}\times\vec{b}) + \beta(\vec{b}\times\vec{c}) + \gamma(\vec{c}\times\vec{a}) = \vec{0}$ (*)

$\Rightarrow \alpha\langle\vec{a}, \vec{a}\times\vec{b}\rangle + \beta\langle\vec{a}, \vec{b}\times\vec{c}\rangle + \gamma\langle\vec{a}, \vec{c}\times\vec{a}\rangle = \langle\vec{a}, \vec{0}\rangle = 0$

$\vec{a}\times\vec{b} \perp \vec{a} \Rightarrow 0$; $\vec{b}\times\vec{c} \perp \vec{a}$ (cross out); $\vec{c}\times\vec{a} \perp \vec{a} = 0$

$= (\vec{a}, \vec{b}, \vec{c})$

(2) $\beta(\vec{a}, \vec{b}, \vec{c}) = 0$ ①

Cum: $\{\vec{a}, \vec{b}, \vec{c}\}$ SLi $\Leftrightarrow$ $\vec{a}, \vec{b}, \vec{c}$ necoplanare

$\Leftrightarrow (\vec{a}, \vec{b}, \vec{c}) \neq 0$ ②

①, ② $\Rightarrow \beta = 0$ ③

Înmulțim scalar în (*) cu $\vec{b}$, vd $\Rightarrow$ asemănător cu $\gamma = 0$ ④

Înmulțim scalar în (*) cu $\vec{c}$, vd $\Rightarrow$ asemănător cu $\alpha = 0$ ⑤

③, ④, ⑤ $\Rightarrow \alpha = \beta = \gamma = 0 \Rightarrow \{\vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}\}$ SLi

„$\Leftarrow$" Fie $\alpha, \beta, \gamma \in \mathbb{R}$ a.î. $\alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = \vec{0}$ (*) $\cdot \times \vec{b}$ $\Rightarrow$

$\Rightarrow \alpha(\vec{a}\times\vec{b}) + \beta(\vec{b}\times\vec{b}) + \gamma(\vec{c}\times\vec{b}) = \vec{0}$

$\Rightarrow \alpha(\vec{a}\times\vec{b}) - \gamma(\vec{b}\times\vec{c}) = \vec{0}$

$\Rightarrow \alpha(\vec{a}\times\vec{b}) - \gamma(\vec{b}\times\vec{c}) + \alpha(\vec{c}\times\vec{a}) = \vec{0}$

dar $\{\vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}\}$ SLi $\Rightarrow$

$\Rightarrow \alpha = \gamma = 0$ (*) $\Rightarrow \beta\vec{b} = \vec{0}$, $\beta \neq 0$ nu $\vec{b} = \vec{0}$ imposibil

pt că dacă $\vec{v} = \vec{0} \Rightarrow \{ \vec{a} \times \vec{b}, \vec{v} \times \vec{c}, (\vec{a}, \vec{b}, \vec{c}) \} = \{ \vec{0}, \vec{0}, (\vec{a}, \vec{b}, \vec{c}) \}$ ≡

Deci $p = 0$ ■

Ex 2: Să se determine ecuația planului P ce conține dreapta $d_p^A = B$ unde $A(-1, 2, 1)$ și $B(1, 1, 2)$.

$d_p^A = B \Rightarrow AB \perp P$

$\vec{AB} = (x_B - x_A)\vec{i} + (y_B - y_A)\vec{j} + (z_B - z_A)\vec{k}$

$= 2\vec{i} - \vec{j} + \vec{k}$

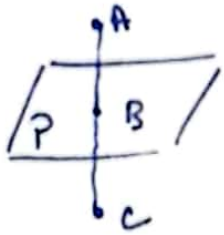
$B \in P$

$P: 2(x-1) + (-1)(y-1) + (z-2) = 0$

$P: 2x - y + z - 2 + 1 - 2 = 0$

$P: 2x - y + z - 3 = 0$ ✓

Ex 3: Să se determine coordonatele simetricului punctului $A(4, -3, 1)$ față de $P: x + 2y - z - 3 = 0$



Obs că $A \notin P$.

Fie $B \in P$ a.î. $AB \perp P$; $\{B\} = \underline{AB} \cap P$; $AB \perp P \Rightarrow$ putem

$\vec{AB} = (x_B - x_A)\vec{i} + (y_B - y_A)\vec{j} + (z_B - z_A)\vec{k}$

$\vec{AB} = (x_B - 4)\vec{i} + (y_B + 3)\vec{j} + (z_B - 1)\vec{k}$

com. vectorul al dr. $\underline{AB}$, un vector normal la P. $\vec{v_{AB}} = \vec{n_P} = 2\vec{i} - \vec{j} + \vec{k}$

$\left.\begin{matrix} AB \perp P \\ B \in P \end{matrix}\right| \Rightarrow P: (x_B - 4)(x - x_B) + (y_B + 3)(y - y_B) + (z_B - 1)(z - z_B) = 0$

$P: x(x_B - 4) + y(y_B + 3)$

$AB: \frac{x-4}{1} = \frac{y+3}{2} = \frac{z-1}{-1} \overset{not}{=} t \Rightarrow$ ec. parametrice ale dreptei AB ▼

$AB: \begin{cases} x = 4 + t \\ y = -3 + 2t \\ z = 1 - t \end{cases}, t \in \mathbb{R}$ (*)

Înlocuind rel. (*) în ec. planului P $\Rightarrow$ t corespunzător punctului B.

$t + 2(-3+2t) - 14t - 3 = 0$

$-6t = 6 \Rightarrow t = -1 \Rightarrow B(5,-1,0)$

$\frac{x_A + x_C}{2} = x_B \Rightarrow x_C = 2x_B - x_A = 10 - 4 = 6$

$\frac{y_A + y_C}{2} = y_B \Rightarrow y_C = 2y_B - y_A = -2 + 3 = 1$

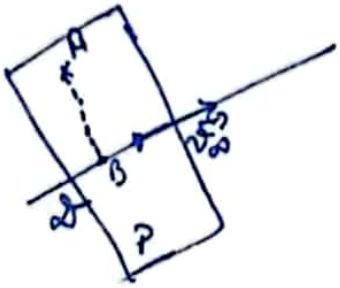
$\parallel \text{—} \Rightarrow z_C = 2z_B - z_A = -1$

$\Rightarrow C(6,1,-1)$

Ex 4 Să se det. coord. simetricului punctului $A(2,3,1)$ față de dr

$d: \frac{x-1}{2} = \frac{y}{-1} = \frac{z-2}{3}$ și $\vec{v}_d = 2\vec{i} - \vec{j} + 3\vec{k}$

obs că $A \notin d$. Fie $B = pr_d^A$



At $\{B\} = d \cap P$, P este planul care conține punctul A și este $\perp d \Rightarrow$

$\vec{N}_P = \vec{v}_d = 2\vec{i} - \vec{j} + 3\vec{k}$

$P: 2(x-2) - (y-3) + 3(z-1) = 0$

$P: 2x - y + 3z - 4 + 3 - 3 = 0$

$P: 2x - y + 3z - 4 = 0$; $\vec{n_P} \perp P$, $A \in P$

$AB: \frac{x-2}{2} = \frac{y-3}{-1} = \frac{z-1}{3} \overset{not}{=} t$ (comun) $\Rightarrow$ ec. dr. $d \Rightarrow 14t = 0$

$\Rightarrow B(-1,0,2)$

$\Rightarrow AB: \begin{cases} x = 2+2t \\ y = 3-t \\ z = 1+3t \end{cases}$ $\Rightarrow 4t + 3 + t + 3 + 9t - 4 = 0$

$14t = 0 \Rightarrow t = 0 \Rightarrow B(2,3,1)$

Fie $C = s_d^A$. Atunci $\frac{x_A + x_C}{2} = x_B \Rightarrow x_C = 2x_B - x_A = -4$

$\Rightarrow y_C = 2y_B - y_A = -3$

$\Rightarrow z_C = 2z_B - z_A = 3$

$\Rightarrow C(-4,-3,3)$

Ex 5: Să se determine ec. planului determinat de dr:

$D_1: \frac{x}{7} = \frac{y+2}{3} = \frac{z-1}{5}$ și $D_2: \frac{x-1}{7} = \frac{y-3}{3} = \frac{z+2}{5}$

a) $\vec{v}_{D_1} = 7\vec{i} + 3\vec{j} + 5\vec{k} = \vec{v}_{D_2}$

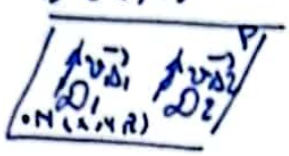
$\Rightarrow D_1 \parallel D_2$ sau coincid $(D_1 \equiv D_2)$

$A(0, -2, 1) \in D_1 \setminus D_2$

$B(1, 3, -2) \in D_2$

$\Rightarrow D_1 \parallel D_2 \Rightarrow$

$\Rightarrow \exists (D_1, D_2)$ planul determinat de dreptele D_1 și D_2



Fie $M(x, y, z) \in P$

$\Leftrightarrow$

$\vec{MA}, \vec{BA}, \vec{v}_{D_1}$ coplanare

$\Leftrightarrow P: (\vec{MA}, \vec{BA}, \vec{v}_{D_1}) = 0 \Leftrightarrow$

$$P: \begin{vmatrix} x_A - x_M & y_A - y_M & z_A - z_M \\ x_A - x_B & y_A - y_B & z_A - z_B \\ 7 & 3 & 5 \end{vmatrix} = 0$$

$$P: \begin{vmatrix} -x_M & -2 - y_M & 1 - z_M \\ -1 & -5 & 3 \\ 7 & 3 & 5 \end{vmatrix} = 0$$

$$P: \begin{vmatrix} x_M & y_M + 2 & z_M - 1 \\ -1 & -5 & 3 \\ 7 & 3 & 5 \end{vmatrix} = 0 \quad (:2)$$

$P: -34x + 26(y+2) + 32(z-1) = 0$

$P: -34x + 26y + 32z + 20 = 0$

Ex 6: Să se stabilească dacă dreptele $D_1: \frac{x-3}{5} = \frac{y+1}{2} = \frac{z-2}{4} = t$ și

$D_2: \frac{x-8}{3} = \frac{y-1}{1} = \frac{z-6}{-2} = s$

sunt coplanare și dacă sunt, să se afle ec. planului det de D_1 și D_2.

$\begin{cases} y = -1+2t \\ z = 2+4t \end{cases}$; $\mathcal{D}_2: \begin{cases} x = 8+3s \\ y = 1+s \\ z = 6-2s \end{cases}$

$\mathcal{D}_1 \cap \mathcal{D}_2 : \begin{cases} 3+5t = 8+3s \\ -1+2t = 1+s \\ 2+4t = 6-2s \end{cases} \Leftrightarrow \begin{cases} 5t-3s = 5 \\ 2t-s = 2 \\ 4t+2s = 4 \\ 2t+s = 2 \end{cases} \Rightarrow t = 1 \Rightarrow s = 0$

$\Rightarrow \mathcal{D}_1 \cap \mathcal{D}_2 = \{A\}$ cu $A(8,1,6) \Rightarrow (\exists)\ P = (\mathcal{D}_1 \mathcal{D}_2)$



$P: \begin{vmatrix} x-x_A & y-y_A & z-z_A \\ 5 & 2 & 4 \\ 3 & 1 & -2 \end{vmatrix} = 0$

$P: \begin{vmatrix} x-8 & y-1 & z-6 \\ 5 & 2 & 4 \\ 3 & 1 & -2 \end{vmatrix} = 0$

$P: -8(x-8) + 22(y-1) - (z-6) = 0$

$P: -8x+22y-z+64-22+6 = 0$

$P: -8x+22y-z+48 = 0$

Ex 7

Să se scrie ecuațiile proiecției dreptei $\mathcal{D}: \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ (d)
pe planul $P: x+y+z-3=0$.

$\mathcal{D}' = pr_P \mathcal{D}$

$A(1,2,3) \in \mathcal{D} \cap P \Rightarrow \mathcal{D} \not\parallel P$ sau $\mathcal{D} \cap P = \{A\}$

$\vec{v}_{\mathcal{D}} \not\perp \vec{N}_P \Leftrightarrow \langle \vec{v}_{\mathcal{D}}, \vec{N}_P \rangle = 0$ (imposibil)

Fie $\mathcal{Q}$ planul ce conține dr $\mathcal{D}$ pt care $\mathcal{Q} \perp P$

(*) $(\exists)\ \mathcal{D}: \begin{cases} \frac{x-1}{2} = \frac{y-2}{3} \\ \frac{y-2}{3} = \frac{z-3}{4} \end{cases} \Leftrightarrow \mathcal{D}: \begin{cases} 3x-2y+1=0 \\ 4y-3z+1=0 \end{cases}$

$\mathcal{D} \subset \mathcal{Q}$

$(\exists)\ \alpha, \beta \in \mathbb{R}$ cu $\alpha^2+\beta^2 \neq 0$ a.î.

$\mathcal{Q}: \alpha(3x-2y+1) + \beta(4y-3z+1) = 0$

$\mathcal{Q}: 3\alpha x + (-2\alpha+4\beta)y - 3\beta z + \alpha + \beta = 0$

$\mathcal{Q} \perp P \Leftrightarrow \vec{N}_{\mathcal{Q}} \perp \vec{N}_P \Leftrightarrow \langle \vec{N}_{\mathcal{Q}}, \vec{N}_P \rangle = 0$

$\vec{N_\beta} = \vec{i} + \vec{j} + \vec{k}$

$\vec{N_\alpha} = 3\alpha\vec{i} + (-2\alpha + 4\beta)\vec{j} - 3\beta\vec{k}$ $\Big|$ $\Leftrightarrow$

$\langle \vec{N_\alpha}, \vec{N_\beta} \rangle = 0$

$\Leftrightarrow 3\alpha - 2\alpha + 4\beta - 3\beta = 0$

$\alpha + \beta = 0 \Rightarrow \underline{\beta = -\alpha}$ $\Big|$ $\Rightarrow 2\alpha^2 \neq 0 \Rightarrow \alpha \neq 0 \Rightarrow \beta \neq 0$

$\alpha^2 + \beta^2 \neq 0$

ec. $\mathcal{Q}$: $\alpha(3x - 2y + 1) - \alpha(4y - 3z + 1) = 0 \;/ : \alpha \neq 0$

$\mathcal{Q}$: $3x - 6y + 3z = 0 \;/ : 3$

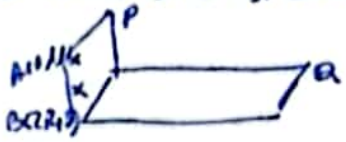
$\mathcal{Q}$: $x - 2y + z = 0$

2) $\mathcal{D}'$: $\begin{cases} x - 2y + z = 0 \\ x + y + z - 3 = 0 \end{cases}$

~~TEMA~~ ~~AED~~ - SS

1) $P = ?$ care conține punctele $A(1,1,1)$ și $B(2,2,3)$ și care este $\perp$ pe α

$\alpha: x+y-z=0$.



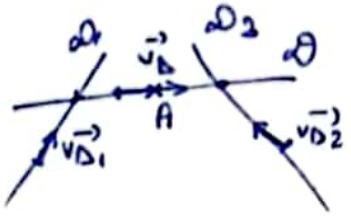
~~$P \perp \alpha \Rightarrow \vec{N_\alpha} = \vec{N_P} = \vec{i} + \vec{j} - \vec{k}$.~~

2) Să se scrie ec. dr. $\mathcal{D}$ care trece prin $A(4,0,-1)$ și $\cap$ dreptele

$\mathcal{D}_1: \frac{x-1}{2} = \frac{y+3}{4} = \frac{z-5}{3} \Rightarrow \vec{v_{D_1}} = 2\vec{i} + 4\vec{j} + 3\vec{k}$

$\mathcal{D}_2: \frac{x}{5} = \frac{y-2}{-1} = \frac{z+1}{2}$

$\Rightarrow \vec{v_{D_2}} = 5\vec{i} - \vec{j} + 2\vec{k}$



$\mathcal{D}: \frac{x-4}{a} = \frac{y}{b} = \frac{z+1}{c}$

$\mathcal{D} = (A, \mathcal{D}_1) \cap (A, \mathcal{D}_2)$

$(\vec{\mathcal{D}}, \vec{\mathcal{D}_1})$ $(\vec{\mathcal{D}}, \vec{\mathcal{D}_2})$

continuare pagina 3

3) dist de la $A(2,3,-1)$ la dr: $\mathcal{D}: \begin{cases} 2x-2y+z+3=0 \\ 3x-2y+2z+17=0 \end{cases}$

4) 68 * dst dist dintre planele paralele

$P: 2x+3y-6z+4=0$ si $Q: 2x+3y-6z-35=0$

---

1) $AB: \frac{x-1}{2-1} = \frac{y-1}{2-1} = \frac{z-1}{3-1}$ (x_A, y_A, z_A)

x_B x_A y_B y_A z_B z_A

$AB: x-1 = y-1 = \frac{z-1}{2}$ (a)

$AB: \begin{cases} x-1 = y-1 \Leftrightarrow x-y=0 \\ x-1 = \frac{z-1}{2} \Leftrightarrow 2x-2 = z-1 \Leftrightarrow 2x-z=1 \Leftrightarrow 2x-z-1=0 \end{cases}$

$\xrightarrow[AB \subset P]{}$ $P: \alpha(x-y) + \beta(2x-z-1) = 0$

$\exists \alpha, \beta \in \mathbb{R}$ cu $\alpha^2+\beta^2 \neq 0$

$\Rightarrow P: (\alpha+2\beta)x - \alpha y - \beta z - \beta = 0$

$P \perp Q \Rightarrow N_P \perp N_Q \Leftrightarrow \langle N_P, N_Q \rangle = 0$

$\vec{N_Q} = \vec{i} + \vec{j} - \vec{k}$

$\Leftrightarrow (\alpha+2\beta) - \alpha + \beta = 0$ ($\alpha^2+\beta^2 \neq 0$)

$3\beta = 0 \Rightarrow \beta = 0 \Rightarrow \alpha \neq 0$

$\Rightarrow P: (\alpha+2\beta)x - \alpha y - \beta z - \beta = 0$

$\Leftrightarrow P: \alpha x - \alpha y = 0 \ / : \alpha \neq 0$

$\Rightarrow P: \underline{x-y=0}$

M II

$\vec{N_Q} = \vec{i} + \vec{j} - \vec{k}$

$\vec{AB} = \vec{i} + \vec{j} + 2\vec{k}$

$A(1,1,1)$

$P: \begin{vmatrix} x-1 & y-1 & z-1 \\ 1 & 1 & -1 \\ 1 & 1 & 2 \end{vmatrix}$

$P: (x-1)(+3) + (y-1)(+4) \cdot 3 + (z-1) \cdot 0 = 0$

$P: 3x-3+3y+3=0$

$P: x-y=0$

$\vec{v}_{D_1} = 2\vec{i} + 4\vec{j} + 3\vec{k}$

$\vec{v}_{D_2} = 5\vec{i} - \vec{j} + 2\vec{k}$

Fie $M(x,y,z) \in (A, D_1) \Leftrightarrow (\vec{AM}, \vec{v}_{D_1}, \vec{AB})$ coplanari.



Fie $B(1,-3,5) \in D_1$ $\qquad A(4,0,-1)$

$(A, D_1): (\vec{AM}, \vec{v}_{D_1}, \vec{AB}) = 0 \Leftrightarrow$

$$(A, D_1): \begin{vmatrix} x-4 & y & z+1 \\ 2 & 4 & 3 \\ -3 & -3 & 6 \end{vmatrix} = 0$$

$(A, D_1): 11(x-4) - 7y + 2(z+1) = 0$

$(A, D_1): 11x - 7y - 2z - 42 = 0$



$M(x,y,z) \in (A, D_2) \Leftrightarrow \vec{AM}, \vec{AC}, \vec{v}_{D_2}$ coplanari

$C(0,2,-1)$

$$\Leftrightarrow (A, D_2): \begin{vmatrix} x-4 & y & z+1 \\ -4 & 2 & 0 \\ 5 & -1 & 2 \end{vmatrix} = 4(x-4) + 8y - 6(z+1)$$

$\Leftrightarrow (A, D_2): 4x + 8y - 6z - 22 = 0$

$$\Rightarrow D: \begin{cases} 11x - 7y - 2z - 42 = 0 \\ 4x + 8y - 6z - 22 = 0 \end{cases}$$

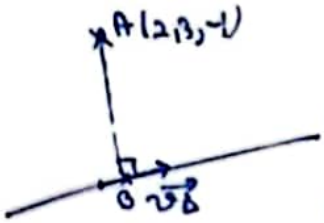
Ex 2) Se cere ec. canonice ale dr D:

$$\vec{v}_D = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 11 & -7 & 2 \\ 4 & 4 & -3 \end{vmatrix} = 13\vec{i} + 37\vec{j} + 58\vec{k}$$

$A(4,0,-1) \in D$

$$\Rightarrow D: \frac{x-4}{13} = \frac{y}{37} = \frac{z+1}{58}$$

3)



Fie $O \in D$ cu $AO \perp D$

Fie P un plan cu $O \in P$ și $P \perp D$

$\vec{v}_D$ coincide cu $\vec{n}_P \Rightarrow P: -2(x-2) + (-1)(y-3) + 2(z+1) = 0$

$$\vec{v}_D = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -2 & 1 \\ 3 & -2 & 2 \end{vmatrix} = -2\vec{i} - 1\cdot\vec{j} + 2\vec{k}$$

$\Rightarrow P: -2x - y + 2z + 4 + 3 + 2 = 0 \Leftrightarrow P: -2x - y + 2z + 9 = 0$

Fie $z = 0 \Rightarrow d: \begin{cases} 2x - 2y = -3 \\ 3x - 2y = -17 \end{cases} \ominus$

$x = -17 + 3 = -14 \Rightarrow -2y = -3 - 2x$

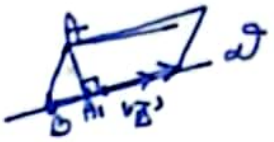
$2y = 2x + 3$

$2y = -28 + 3$

$y = -\frac{25}{2}$

$\Rightarrow d: \frac{x+14}{-2} = \frac{y + \frac{25}{2}}{-1} = \frac{z}{2} = t$

$\Rightarrow \begin{cases} x = -14 - 2t \\ y = -25/2 - t \\ z = 2t \end{cases} \rightsquigarrow_P \Rightarrow \dots$



Mat prof Fie $D(-14, -25/2, 0)$

Fie $A_1 = pr_d A$. Atunci $d(A, d) = AA_1 = \|\vec{AA_1}\|$

$\|\vec{AB} \times \vec{v_D}\| = d(A,B) \cdot \|\vec{v_D}\| \Rightarrow d(A, d) = \frac{\|\vec{AB} \times \vec{v_D}\|}{\|\vec{v_D}\|}$

$\vec{AB} = -16\vec{i} - \frac{31}{2}\vec{j} + \vec{k}$

$\vec{v_D} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -2 & 1 \\ 3 & -2 & 2 \end{vmatrix} = -2\vec{i} - \vec{j} + 2\vec{k} \Rightarrow \|\vec{v_D}\| = 3$

$\vec{AB} \times \vec{v_D} = \frac{1}{2} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -32 & -31 & 2 \\ -2 & -1 & 2 \end{vmatrix} = \frac{1}{2}(-60\vec{i} + 60\vec{j} - 30\vec{k})$

$= \frac{1}{2}(-30(2i - 2\vec{j} + \vec{k}))$

$\Rightarrow \|\vec{AB} \times \vec{v_D}\| = 45$

$\Rightarrow d(A, d) = \frac{45}{3} = 15.$

4) $d(P, Q)$

$P: 2x + 3y - 6z + 14 = 0$

$Q: 2x + 3y - 6z - 35 = 0$

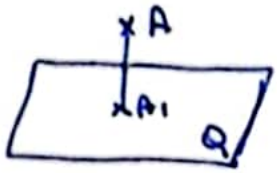
Sol Met I

Fie $A(-7, 0, 0)$. $d(P, Q) = \min_{\substack{N \in P \\ N \in Q}} d(M, N) \underset{P \parallel Q}{=} d(A, Q) = \frac{|2 \cdot (-7) + 3 \cdot 0 - 35|}{\sqrt{4 + 9 + 36}}$

$= \frac{49}{\sqrt{49}} = 7$

Met II

Fie $A_1 = pr_Q A$. Atunci $AA_1 = d(A,Q)$



$AA_1: \frac{x+7}{2} = \frac{y}{3} = \frac{z}{-6} = \lambda$

$AA_1 \perp Q \Rightarrow \vec{v}_{AA_1} = \vec{N_Q} = 2\vec{i} + 3\vec{j} - 6\vec{k}$

$AA_1: \begin{cases} x = -7+2s \\ y = 3s \\ z = -6s \end{cases}$ ∩ Q

$AA_1 \cap Q = \{A_1\}$

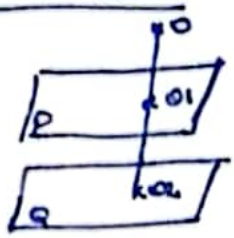
$\Rightarrow 2(-7+2s) + 9s + 36s - 35 = 0$

$49s = 49 \Rightarrow s = 1$

$\Rightarrow A_1(-5, 3, -6)$

$\Rightarrow \|AA_1\| = \sqrt{4+9+36} = \sqrt{49} = 7.$

Met a III-a



$d(O,P_1) = \frac{14}{7}$

$d(O,Q_1) = \frac{35}{7}$

$\Rightarrow d(P,Q) = \frac{35}{7} - \frac{14}{7} =$

(NU real)

(pentru originea este între plane)

5) Aplicații liniare

Fie $A = \begin{pmatrix} 1 & 2 \\ 3 & 6 \\ -2 & -4 \end{pmatrix}$ și $f: \underbrace{\mathcal{M}_2(\mathbb{R})}_{\text{sp. vect. real}} \to \mathcal{M}_{3,2}(\mathbb{R})$

$f(X) = A \cdot X, \ \forall X \in \mathcal{M}_2(\mathbb{R})$

Să se arate f apl. lin și să se det o bază în Ker f și în Im f.

f apl. liniară $\Leftrightarrow f(\alpha B + \beta C) = \alpha f(B) + \beta f(C) \quad \forall \alpha, \beta \in \mathbb{R}$ și $B, C \in \mathcal{M}_2(\mathbb{R})$

$f(\alpha B+\beta C) = A(\alpha B+\beta C) = \alpha AB+\beta AC$

$\alpha f(B)+\beta f(C) = \alpha AB+\beta AC$ $\Big| \Rightarrow f$ apl. liniară

$\operatorname{Im} f \overset{\text{def}}{=} \{\underbrace{f(X)}_{AX} \mid X \in \mathcal{M}_2(\mathbb{R})\}. \subseteq \mathcal{M}_{3,2}(\mathbb{R})$

$\operatorname{Im} f = \{ A\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a,b,c,d \in \mathbb{R}\} =$

$= \left\{ \begin{pmatrix} 1 & 2 \\ 3 & 6 \\ -2 & -4 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid \dots \right\} = \left\{ \begin{pmatrix} a+2c & b+2d \\ 3a+6c & 3b+6d \\ -2a-4c & -2b-4d \end{pmatrix} \mid a,b,c,d \in \mathbb{R} \right\}$

$= \left\{ a\begin{pmatrix} 1 & 0 \\ 3 & 0 \\ -2 & 0 \end{pmatrix} + b\begin{pmatrix} 0 & 1 \\ 0 & 3 \\ 0 & -2 \end{pmatrix} + c\begin{pmatrix} 2 & 0 \\ 6 & 0 \\ -4 & 0 \end{pmatrix} + 2d\begin{pmatrix} 0 & 1 \\ 0 & 3 \\ 0 & -2 \end{pmatrix} \Big| a,b,c,d \in \mathbb{R} \right\}$

$2c\begin{pmatrix} 1 & 0 \\ 3 & 0 \\ -2 & 0 \end{pmatrix}$

$\Rightarrow \operatorname{Im} f = \operatorname{sp}\left\{ \begin{pmatrix} 1 & 0 \\ 3 & 0 \\ -2 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 3 \\ 0 & -2 \end{pmatrix} \right\}$

dar $\left\{ \begin{pmatrix} 1 & 0 \\ 3 & 0 \\ -2 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 3 \\ 0 & -2 \end{pmatrix} \right\}$ SLi $\Big| \Rightarrow \left\{ \begin{pmatrix} 1 & 0 \\ 3 & 0 \\ -2 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 3 \\ 0 & -2 \end{pmatrix} \right\}$ baza pt $\operatorname{Im} f$ și $\dim_{\mathbb{R}} \operatorname{Im} f = 2$

t.
rang-def $\Rightarrow \dim_{\mathbb{R}} \operatorname{Ker} f = \dim_{\mathbb{R}} \mathcal{M}_2(\mathbb{R}) - \dim_{\mathbb{R}} \operatorname{Im} f = 2$

$\operatorname{Ker} f = \{ X \in \mathcal{M}_2(\mathbb{R}) \mid f(X) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \}$

$= \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathcal{M}_2(\mathbb{R}) \mid a+2c = b+2d = 0 \right\}$

$= \left\{ \begin{pmatrix} -2c-2d \\ c \quad d \end{pmatrix} \mid c,d \in \mathbb{R} \right\}$

$= \left\{ c\begin{pmatrix} -2 & 0 \\ 1 & 0 \end{pmatrix} + d\begin{pmatrix} 0 & -2 \\ 0 & 1 \end{pmatrix} \mid c,d \in \mathbb{R} \right\}$

$\Rightarrow \left\{ \begin{pmatrix} -2 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -2 \\ 0 & 1 \end{pmatrix} \right\}$ SG ($\operatorname{Ker} f = \operatorname{sp}\left\{ \begin{pmatrix} -2 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -2 \\ 0 & 1 \end{pmatrix} \right\}$)

dar $\dim_{\mathbb{R}} \operatorname{Ker} f = 2$

$\Rightarrow \left\{ \begin{pmatrix} -2 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -2 \\ 0 & 1 \end{pmatrix} \right\}$ o bază pt $\operatorname{Ker} f$.

~~Algebră~~

1) Fie V un $\mathbb{K}$-sp vect cu $\dim_{\mathbb{K}} V = 5$ și $f: V \to V$ op. liniară $\Rightarrow \operatorname{Ker} f, \operatorname{Im} f \subseteq V$

Dacă $\dim_{\mathbb{K}}(\operatorname{Ker} f \cap \operatorname{Im} f) = 3$. Să se dem.

că $\operatorname{Ker} f \subseteq \operatorname{Im} f$ sau $\operatorname{Ker} f \supseteq \operatorname{Im} f$. Este posibil ca $\dim_{\mathbb{K}} \operatorname{Ker} f \cap \operatorname{Im} f = 3$?

T1: $f: V \to V$ apl. lin. $\dim_{\mathbb{K}} V < +\infty$ atunci

$\dim_{\mathbb{K}} V = \dim_{\mathbb{K}} \operatorname{Ker} f + \dim_{\mathbb{K}} \operatorname{Im} f$

P1: Dacă V este un $\mathbb{K}$-sp-vect finit dim și $U \underset{\text{sp.vect.}}{\subseteq} V$ at:

i) $\dim_{\mathbb{K}} U \leq \dim_{\mathbb{K}} V$

ii) $\dim_{\mathbb{K}} U = \dim_{\mathbb{K}} V \Rightarrow U = V$

T2 (Grassmann) Fie W un $\mathbb{K}$-spațiu vectorial finit dimensional. $U, V \underset{\text{subsp. vect.}}{\subseteq} W$ at

$$\dim_{\mathbb{K}} U + \dim_{\mathbb{K}} V = \dim_{\mathbb{K}}(U+V) + \dim_{\mathbb{K}}(U \cap V)$$

Obs. $U + V \overset{\text{def}}{=} \{u + v \mid u \in U, v \in V\}$

$$\left.\begin{matrix} T_1 \\ T_2 \end{matrix}\right| \Rightarrow \underset{\uparrow \dim_{\mathbb{K}} V}{5} = \underset{\uparrow T_1}{\dim_{\mathbb{K}} \operatorname{Ker} f + \dim_{\mathbb{K}} \operatorname{Im} f} \overset{T_2}{=} \dim_{\mathbb{K}}(\operatorname{Ker} f + \operatorname{Im} f) + \dim_{\mathbb{K}}(\operatorname{Ker} f \cap \operatorname{Im} f)$$

$$\Rightarrow \dim_{\mathbb{K}}(\operatorname{Ker} f + \operatorname{Im} f) = 3$$

$$\operatorname{Ker} f \cap \operatorname{Im} f \subseteq \operatorname{Ker} f \subseteq \operatorname{Ker} f + \operatorname{Im} f \overset{P_{1i)}}{\Rightarrow}$$

$$2 \leq \dim_{\mathbb{K}} \operatorname{Ker} f \leq 3 \Rightarrow \dim_{\mathbb{K}} \operatorname{Ker} f \in \{2, 3\}.$$

I Dacă $\dim_{\mathbb{K}} \operatorname{Ker} f = 2$

$\operatorname{Ker} f \cap \operatorname{Im} f \subseteq \operatorname{Ker} f$

$\dim_{\mathbb{K}}(\operatorname{Ker} f \cap \operatorname{Im} f) = 2$

$\Rightarrow \operatorname{Ker} f = \operatorname{Ker} f \cap \operatorname{Im} f \subseteq \operatorname{Im} f$

I Dacă $\dim_{\mathbb{K}} \operatorname{Ker} f = 3 \Rightarrow \dim_{\mathbb{K}} \operatorname{Im} f = 2 \overset{\text{def}}{\Rightarrow} \operatorname{Im} f = \operatorname{Ker} f \cap \operatorname{Im} f \subseteq \operatorname{Ker} f$

Dacă $\dim(\operatorname{Ker} f \cap \operatorname{Im} f) = 3 \overset{\text{T2}}{\Rightarrow} \dim(\operatorname{Ker} f + \operatorname{Im} f) = 2$

, dar $\operatorname{Ker} f \cap \operatorname{Im} f \subset \operatorname{Ker} f + \operatorname{Im} f \Rightarrow \dim(\ldots \cap \ldots) \leq \dim(\ldots + \ldots)$ ↯

Ex 2: Fie $f: \mathbb{R}^4 \to \mathcal{M}_2(\mathbb{C})$ (sp. vect. real → sp. vect. real), $f(x) = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}$

, unde $x \overset{not}{=} (x_1, x_2, x_3, x_4) \in \mathbb{R}^4$

$a \overset{not}{=} x_1 + ix_2$ și $b \overset{not}{=} x_3 + ix_4$ $\Rightarrow$

$\Rightarrow f(x_1, x_2, x_3, x_4) = \begin{pmatrix} x_1 + ix_2 & x_3 + ix_4 \\ -x_3 + ix_4 & x_1 - ix_2 \end{pmatrix}$

Să se arate că f aplicație liniară (temă). Să se scrie matr. lui f rel la bazele canonice. Să se det câte câte câte pt $\operatorname{Im} f$ și $\operatorname{Ker} f$ ~~găsim~~.

$B_1 \overset{not}{=} (B_c)_{\mathbb{R}^4} = \{e_1 = (1,0,0,0), e_2 = (0,1,0,0), e_3 = (0,0,1,0), e_4 = (0,0,0,1)\}$

$B_2 \overset{not}{=} (B_c)_{\mathcal{M}_2(\mathbb{C})} = \left\{\bar{\varepsilon}_1, i\bar{\varepsilon}_1, \bar{\varepsilon}_2, i\bar{\varepsilon}_2, \bar{\varepsilon}_3, i\bar{\varepsilon}_3, \bar{\varepsilon}_4, i\bar{\varepsilon}_4\right\}$

$\bar{\varepsilon}_1 \overset{not}{=} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \bar{\varepsilon}_2 \overset{not}{=} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \bar{\varepsilon}_3 \overset{not}{=} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \bar{\varepsilon}_4 \overset{not}{=} \begin{pmatrix} 0 & i \\ 0 & 0 \end{pmatrix}$

$\dim_{\mathbb{R}} \mathcal{M}_2(\mathbb{C}) = 8$

! $\dim_{\mathbb{R}} \mathcal{M}_2(\mathbb{R}) = 4$

$f(e_1) = f(1,0,0,0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \bar{\varepsilon}_1 + \bar{\varepsilon}_4$; $f(e_2) = f(0,1,0,0) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$

$f(e_3) = f(0,0,1,0) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$; $f(e_4) = f(0,0,0,1) = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$

$M_{B_1,B_2}(f) = \begin{pmatrix} f(e_1) & f(e_2) & f(e_3) & f(e_4) \\ 1 & 0 & 0 & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & i \end{pmatrix}$

$f(e_1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \bar{\varepsilon}_1 + \bar{\varepsilon}_4 = 1 \cdot \bar{\varepsilon}_1 + 0 \cdot i\bar{\varepsilon}_1 + 0 \cdot \bar{\varepsilon}_2 + 0 \cdot i\bar{\varepsilon}_2 + 0 \cdot \bar{\varepsilon}_3 + 0 \cdot i\bar{\varepsilon}_3 + 1 \cdot \bar{\varepsilon}_4 + 0 \cdot i\bar{\varepsilon}_4$

$\ker f = \{x \in \mathbb{R}^4 \mid f(x) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\} =$

$= \{(x_1,x_2,x_3,x_4) \in \mathbb{R}^4 \mid \begin{pmatrix} x_1+ix_2 & x_3+ix_4 \\ -x_3+ix_4 & x_1-ix_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\}$

$\Leftrightarrow$

$x_1+ix_2 = 0 \overset{\text{def}}{\Leftrightarrow}$ $\begin{cases} x_1 = x_2 = 0 \\ x_3 = x_4 = 0 \end{cases}$

$x_3+ix_4 = 0$ pentru orice

$x_1, x_2, x_3, x_4 \in \mathbb{R}$

$\Rightarrow \ker f = \{(0,0,0,0)\} \Rightarrow f$ injectivă

$\Rightarrow \dim_{\mathbb{R}} \operatorname{Im} f = 4 = \dim_{\mathbb{R}} \mathbb{R}^4$ ①

!! $\dim_{\mathbb{K}} V = 0 \Leftrightarrow V = \{0_V\}$

$\operatorname{Im} f = \{f(x_1,x_2,x_3,x_4) \mid x_1,\dots,x_4 \in \mathbb{R}\}$

$= \{x_1 f(e_1) + x_2 f(e_2) + \dots + x_4 f(e_4) \mid x_1,\dots,x_4 \in \mathbb{R}\}$

$\operatorname{Im} f = \langle f(e_1),\dots,f(e_4)\rangle$ ②

Din ①, ② $\Rightarrow$ o bază pt $\operatorname{Im} f$ este subm. $\{f(e_1),\dots,f(e_4)\}$

Dacă $\dim_{\mathbb{K}} V = n$ și $\{v_1,\dots,v_n\} \subseteq V$

atunci $\{v_1,\dots,v_n\}$ este bază pt $V \Leftrightarrow \{v_1,\dots,v_n\} \subseteq_{SG} V$

$\Leftrightarrow \{v_1,\dots,v_n\}$ sli.

Ex 3. Fie $f: \mathbb{R}^3 \to \mathbb{R}^3$ apl. liniară a.î. $f(0,1,0) = (3,8,2)$,

$f(-2,1,1) = (2,7,4)$

$f(5,1,-3) = (5,8,-5)$. Să se arate

că f izomorfism de sp. vect și să se det f^{-1}.

$f(\underbrace{x,y,z}_{\in \mathbb{R}^3}) = f(\underbrace{x}_{1,0,0)} e_1 + \underbrace{y}_{1} e_2 + z e_3) = x f(e_1) + y \underbrace{f(e_2)}_{(3,8,2)} + z f(e_3)$

$f(-2,1,1) = (2,7,4) \Leftrightarrow -2f(e_1) + f(e_3) = (2,7,4) - (3,8,2)$

$f(5,1,-3) = (5,8,-5) \Leftrightarrow 5f(e_1) - 3f(e_3) = (5,8,-5) - (3,8,2)$

$\Rightarrow$ 1) $\begin{cases} -2f(e_1) + f(e_3) = (-1,-1,2) \mid \cdot 3 \\ 5f(e_1) - 3f(e_3) = (2,0,-7) \end{cases}$

2) $\begin{cases} -6f(e_1) + 3f(e_3) = (-3,-3,6) \\ 5f(e_1) - 3f(e_3) = (2,0,-7) \end{cases}$

"-" $\Rightarrow f(e_1) = (-1,-3,13) \Rightarrow f(e_1) = (1,3,-13)$

$f(e_3) = (-1,-1,2) + (2,6,4) = (1,5,4)$

a) $f(x,y,z) = x(1,3,1) + y(3,8,2) + z(1,5,4)$

$f(x,y,z) = (x+3y+z)(1,0,0),\ 3x+8y+5z,\ x+2y+4z)$

$$M_{B_C}(f) = \begin{pmatrix} f(e_1) & f(e_2) & f(e_3) \\ 1 & 3 & 1 \\ 3 & 8 & 5 \\ 1 & 2 & 4 \end{pmatrix} \overset{not}{=} A$$

Dacă $\det A \neq 0 \overset{curs}{\Rightarrow} f$ izomorfism

$$\det A = 32 + 6 + 15 - 8 - 10 - 36$$
$$= -1 \neq 0 \Rightarrow (\exists) A^{-1} \text{ si } (\exists) f^{-1}, \text{ ion}$$

$$M_{B_C}(f^{-1}) = [M_{B_C}(f)]^{-1} = A^{-1}$$

$$\left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 3 & 8 & 5 & 0 & 1 & 0 \\ 1 & 2 & 4 & 0 & 0 & 1 \end{array}\right) \xrightarrow[L_3 \to L_3 - L_1]{L_2 \to L_2 - 3L_1} \left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & -1 & 2 & -3 & 1 & 0 \\ 0 & -1 & 3 & -1 & 0 & 1 \end{array}\right) \xrightarrow{L_3 \to L_3 - L_2} \left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & -1 & 2 & -3 & 1 & 0 \\ 0 & 0 & 1 & 2 & -1 & 1 \end{array}\right)$$

$$\xrightarrow[L_1 \to L_1 - L_3]{L_2 \to L_2 - 2L_3} \left(\begin{array}{ccc|ccc} 1 & 3 & 0 & -1 & 1 & -1 \\ 0 & -1 & 0 & -7 & 3 & -2 \\ 0 & 0 & 1 & 2 & -1 & 1 \end{array}\right) \xrightarrow{L_1 \to L_1 + 3L_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -22 & 10 & -7 \\ 0 & 1 & 0 & 7 & -3 & 2 \\ 0 & 0 & 1 & 2 & -1 & 1 \end{array}\right)$$

$f^{-1}(e_1)\ f^{-1}(e_2)\ f^{-1}(e_3)$

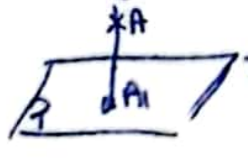
$f^{-1}(x,y,z) = x f^{-1}(e_1) + y f^{-1}(e_2) + z f^{-1}(e_3)$

$= x(-22,7,2) + y(10,-3,-1) + z(-7,2,1)$

$= (-22x + 10y - 7z,\ 7x - 3y + 2z,\ 2x - y + z)$

Ex4: Fie $f: \mathbb{R}^2 \to \mathbb{R}^3$

$f(u,v) =$ „proiecția punctului de coord $(u,v,0)$ pe planul $P: 3y - y - 2z = 0$". Să se arate că f este apl. liniară și să se det. câte o bază pt $\operatorname{Im} f$, resp $\operatorname{Ker} f$.



Fie $A_1 = \pi_P A$

At AA_1: $\frac{x-u}{3} = \frac{y-v}{-1} = \frac{z}{-2}$

$A_1: \begin{cases} x = u+3t \\ y = v-t \\ z = -2t \end{cases}, t \in \mathbb{R}$

$\Rightarrow 3u+9t-v+t+4t=0 \Rightarrow 14t = v-3u$

$$t = \frac{v-3u}{14}$$

$$2) \begin{cases} x_{A_1} = u + \frac{3}{14}(v-3u) = \frac{1}{14}(5u+3v) \\ y_{A_1} = v - \frac{1}{14}(v-3u) = \frac{1}{14}(3u+13v) \\ z_{A_1} = -\frac{2}{14}(v-3u) = \frac{1}{14}(6u-2v) \end{cases} \Rightarrow$$

$$\Rightarrow f(u,v) = \frac{1}{14}(5u+3v, 3u+13v, 6u-2v)$$

$\operatorname{Ker} f = \{(x,y) \in \mathbb{R}^2 \mid 5x+3y=0, 3x+13y=0, 6x-2y=0\}$

$= \{(0,0)\} \Rightarrow f$ inj. Și $\dim_{\mathbb{R}} \operatorname{Im} f = 2 \neq 3 = \dim_{\mathbb{R}} \mathbb{R}^3$

$\Rightarrow f$ nu e surj.

Ex 5: Fie $f: \mathbb{R}^3 \to \mathbb{R}^3$, $f(x,y,z) = (2x-4y+2z, 2z, -2x+6y+z)$

Definim $g: \mathbb{R}^3 \to \mathbb{R}^3$, $g = f^2 + 2f$

$(f^2 \stackrel{not}{=} f \circ f)$

Să se det. câte o bază și $\dim$ pt $\operatorname{Ker} g$, $\operatorname{Im} g$

Fie $A = M_{B_c}(f) = \begin{pmatrix} 2 & -4 & 2 \\ 0 & 0 & 2 \\ -2 & 6 & 1 \end{pmatrix}$!! $\Rightarrow M_{B_c}(g) = A^2 + 2A$

$f(e_1) \dots$

$$A^2 = \begin{pmatrix} 2 & -4 & 2 \\ 0 & 0 & 2 \\ -2 & 6 & 1 \end{pmatrix}\begin{pmatrix} 2 & -4 & 2 \\ 0 & 0 & 2 \\ -2 & 6 & 1 \end{pmatrix} = \begin{pmatrix} -4 & 4 & -2 \\ -4 & 12 & 2 \\ -6 & 14 & 3 \end{pmatrix}$$

$$\Rightarrow A^2 + A = \begin{pmatrix} -2 & 0 & 0 \\ -4 & 12 & 4 \\ -8 & 20 & 10 \end{pmatrix}$$

$g(e_1)$ $g(e_2)$ $g(e_3)$

$\Rightarrow g(x,y,z) = (-2x-4x+12y+4z, -8x+20y+10z)$

$\operatorname{Ker} g = \{(x,y,z) \mid g(x,y,z) = 0\}$

$\Rightarrow \operatorname{Ker} g = \{(x,y,z) \mid \begin{cases} -2x = 0 \\ -4x+12y+4z=0 \\ -8x+20y+10z=0 \end{cases}\} \Rightarrow \dots \Rightarrow \operatorname{Ker} g = \{(0,0,0)\}$

$\Rightarrow \dim_{\mathbb{R}} \operatorname{Ker} g = 0$

$\Rightarrow \dim_{\mathbb{R}} \operatorname{Im} g = 3$

$\operatorname{Im} g \subseteq_{\text{subspace}} \mathbb{R}^3$

$\dim_{\mathbb{R}} \mathbb{R}^3 = 3 \Rightarrow$

2) $\mathrm{Im}\, g = \mathbb{R}^3$

ALGEB - Aplicații liniare

Def: Fie V, W două $\mathbb{K}$-sp vect și $f: V \to W$.
f s.n. aplicație liniară $\Leftrightarrow$

→ $(\forall) v_1, v_2 \in V$; $f(v_1 + v_2) = f(v_1) + f(v_2)$

→ $(\forall) v_1, v_2 \in V$ și $(\forall) \alpha, \beta \in \mathbb{K} \Rightarrow f(\alpha v_1 + \beta v_2) = \alpha f(v_1) + \beta f(v_2)$

- $V = W \Rightarrow$ ENDOMORFISM
- f bijectivă $\Rightarrow$ izomorfism
  - $V = W \Rightarrow$ AUTOMORFISM

Prop: → $f: V \to W$ izomorfism $\Rightarrow$
$f^{-1}: W \to V$ izomorfism

→ U, V, W $\mathbb{K}$-sp vect.
$U \xrightarrow{f} V \xrightarrow{g} W$
f, g izomorfisme
$\Rightarrow (g \circ f): U \to W$ izomorfism

Teorema de izomorfism

→ V un $\mathbb{K}$-spvect cu $\dim_{\mathbb{K}} V = n$. Atunci $V \cong \mathbb{K}^n$

→ V și W $\mathbb{K}$-spved. $V \cong W \Leftrightarrow \dim_{\mathbb{K}} V = \dim_{\mathbb{K}} W$

! $\operatorname{Ker} f \overset{def}{=} \{ v \in V \mid f(v) = 0_W \} \subseteq V$
! $\operatorname{Im} f \overset{def}{=} \{ f(v) \mid v \in V \} \subseteq W$
; $f: V \to W$, ap. lin.

PROPRIETĂȚI: Ker f și Im f

i) $\operatorname{Ker} f \underset{\text{subsp vect}}{\subseteq} V$; $\operatorname{Im} f \underset{\text{subsp vect}}{\subseteq} W$

ii) f inj $\Leftrightarrow \operatorname{Ker} f = \{0_V\}$; ! $\dim_{\mathbb{K}} \operatorname{Ker} f = 0$

iii) f surj $\Leftrightarrow \operatorname{Im} f = W$

iv) Teorema rang-defect

Dacă $\dim_{\mathbb{K}} V < +\infty$, atunci

$$\boxed{\dim_{\mathbb{K}} V = \dim_{\mathbb{K}} \ker f + \dim_{\mathbb{K}} \operatorname{Im} f}$$

v) Dacă $\dim_{\mathbb{K}} V = \dim_{\mathbb{K}} W < +\infty$, at [$f$ inj $\Leftrightarrow$ f surj]

TEOREMA GRASSMANN

Fie W un $\mathbb{K}$-spvect finit dim. $U, V \underset{\text{subsp vect}}{\subseteq} W$

$$\Rightarrow \dim_{\mathbb{K}} U + \dim_{\mathbb{K}} V = \dim_{\mathbb{K}} (U+V) + \dim_{\mathbb{K}} (U \cap V)$$

Prop: Fie U, V, W 3 $\mathbb{K}$-spvect. cu $\dim_{\mathbb{K}} U = n$, $\dim_{\mathbb{K}} V = m$ și $\dim_{\mathbb{K}} W = p$; $f: U \to V$, $g: V \to W$ | ap. lin.

Dacă B_1 bază U, $B_2 \to V$, $B_3 \to W$, atunci pentru $(g \circ f): U \to W$, avem:

$$M_{B_1, B_3}^{(g \circ f)} = M_{B_2, B_3}^{(g)} \cdot M_{B_1 B_2}^{(f)}$$

Prop: → Fie V un $\mathbb{K}$-spvect finit gen, $\dim_{\mathbb{K}} V < +\infty$. B bază pt V. $f: V \to V$ ap. lin cu $n \in \mathbb{N}^*$. Atunci:

$$M_B(f^n) = [M_B f]^n, \quad f^n = \underbrace{f \circ f \circ \dots \circ f}_{\text{de } n \text{ ori}}$$

→ $\dim_{\mathbb{K}} V = m \Rightarrow M_B(1_V) = I_m$

! $1_V: V \to V$; $1_V(v) = v$

Fie: U, V 2 $\mathbb{K}$-sp vect cu $\dim_{\mathbb{K}} U = \dim_{\mathbb{K}} V = n$;
$f: U \to V$ ap lin:

→ f izomorfism $\Leftrightarrow$ $\{\det(M_{B_1,B_2}(f)) \neq 0\}$, $(\forall) B_1$ bază pt U / $(\forall) B_2$ bază pt V

→ f izomorfism → $\{(\forall) B_1$ bază pt U, $(\forall) B_2$ bază pt V $\Rightarrow M_{B_2,B_1}(f^{-1}) = [M_{B_1,B_2}(f)]^{-1}\}$

Def: V un $\mathbb{K}$-spvect finit gen și B_1, B_2 2 baze pt V.
S.n. MATRICE DE TRECERE de la baza B_1 la B_2 și se notează $M_{B_1 \to B_2}$ i matricea alpecărei coloane ne află coord. vectorilor din B_2 în bază B_1.

! $\{M_{B_1 \to B_2} = M_{B_2,B_1}({}^{t}V)\}$

Prop: $f: U \to V$ ap. lin. U, V sp. vect finit dim cu B_1 bază pt U și B_2 bază pt V, atunci $(\forall) u \in U$:

$\{[f(u)]_{B_2} = M_{B_1,B_2}(f) \cdot [u]_{B_1}\}$

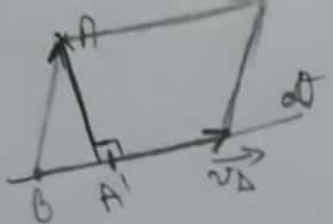
coloana coordonatelor lui $f(u)$ în baza B_2 ↑ — ↑ coloana coordonatelor lui u în baza B_1

Prop: V $\mathbb{K}$-spvect, $\dim_{\mathbb{K}} V < +\infty$. B_1, B_2 baze pt V.
Atunci: $(\forall) v \in V$, $\{[v]_{B_1} = M_{B_1 \to B_2} \cdot [v]_{B_2}\}$.

TEOREMĂ

U, V $\mathbb{K}$-spvect, finit dim. B_1, B_1' baze pt U; B_2, B_2' baze pt V.
$f: U \to V$ ap. lin. Not $A \overset{not}{=} M_{B_1 \to B_1'}$, $C \overset{not}{=} M_{B_2 \to B_2'}$

$\Rightarrow \{M_{B_1',B_2'}(f) = C^{-1} \cdot M_{B_1,B_2} \cdot A\}$



Fie $B \in d$ și $\vec{v}_D$ vectorul director al lui Δ.

În paralelogramul obținut, ∞ obs.

$\mathcal{A} = \|\vec{v}_\Delta\| \cdot AA'$, $AA' \perp \Delta$.
(aria) $\quad\quad$ $AA' = d(A, \Delta)$

$$\Downarrow$$

$$\|\vec{AB} \times \vec{v}_\Delta\| = \|\vec{v}_\Delta\| \cdot AA'$$

$$\Rightarrow AA' = d(A, \Delta) = \frac{\|\vec{AB} \times \vec{v}_\Delta\|}{\|\vec{v}_\Delta\|} \quad (1)$$

(B ∞ alege convenabil) $\Rightarrow$ ex. lui $\vec{AB} = \alpha_1\vec{i} + \alpha_2\vec{j} + \alpha_3\vec{k}$, $\vec{v}_\Delta = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$

$$\Rightarrow \vec{AB} \times \vec{v}_\Delta = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \alpha_1 & \alpha_2 & \alpha_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \beta_1\vec{i} + \beta_2\vec{j} + \beta_3\vec{k}$$

$$\Rightarrow \|\vec{AB} \times \vec{v}_\Delta\| = \sqrt{\beta_1^2 + \beta_2^2 + \beta_3^2} \quad (2)$$

$$\|\vec{v}_\Delta\| = \sqrt{v_1^2 + v_2^2 + v_3^2} \quad (3)$$

(1), (2), (3) $\Rightarrow d(A, \Delta) = \dfrac{\sqrt{\beta_1^2 + \beta_2^2 + \beta_3^2}}{\sqrt{v_1^2 + v_2^2 + v_3^2}}$ //

---

1 (2) PARȚIAL 2021 (practic) $\quad f\begin{pmatrix} 6 & x \\ 8 & 9 \end{pmatrix} = \begin{pmatrix} 6x \\ 8 \end{pmatrix} = f\left(6 \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + x\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\right)$

$f : \mathcal{M}_2(\mathbb{R}) \to \mathbb{R}^4$; $\operatorname{Ker} f = U$.

$\operatorname{Ker} f = U \Leftrightarrow f\begin{pmatrix} 6 & x \\ 8 & 9 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$; $f\begin{pmatrix} 3 & 2 \\ 9 & 8 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

$$\begin{cases} 6f(e_1) + xf(e_2) + 8f(e_3) + 9f(e_4) = 0_{\mathbb{R}^4} \\ 3f(e_1) + 2f(e_2) + 9f(e_3) + 8f(e_4) = 0 \end{cases}$$

$f(e_3) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

$f(e_4) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$, alese convenabil.

$$\begin{cases} 6f(e_1) + xf(e_2) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ 3f(e_1) + 2f(e_2) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad / \cdot(-2) \end{cases} \Rightarrow 3f(e_1) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} - 2f(e_2)$$

$$\Leftrightarrow \begin{cases} 6f(e_1) + xf(e_2) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ -6f(e_1) - 4f(e_2) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \end{cases} \Rightarrow 3f(e_1) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} - 2f(e_2) = \begin{pmatrix} 0 \\ 0 \\ 10/3 \\ 11/3 \end{pmatrix}$$

$V_1 = \{P, Q, R\}$ $P = 1 - X + 2X^2 + 4X^3 + 3X^4$, $Q = -1 + 2X + X^2 + X^4$, $R = -2 + 5X + 7X^2 + 6X^3 + 6X^4$.

$$V_1 \begin{pmatrix} P & Q & R \\ 1 & -1 & -2 \\ -1 & 2 & 5 \\ 2 & 1 & 5 \\ 4 & 0 & 4 \\ 3 & 1 & 6 \end{pmatrix} \xrightarrow[L_3 \to L_3 - 2L_1 \\ L_4 \to L_4 - 4L_1 \\ L_5 \to L_5 - 3L_1]{L_2 \to L_2 + L_1} \begin{pmatrix} 1 & -1 & -2 \\ 0 & 1 & 3 \\ 0 & 3 & 9 \\ 0 & 4 & 12 \\ 0 & 4 & 12 \end{pmatrix} \xrightarrow[L_4 \to L_4 - L_4 \\ L_5 \to L_5 - L_5]{L_3 \to L_3 - L_3} \begin{pmatrix} 1 & -1 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{matrix} \\ \\ \\ \\ \\ P \end{matrix} \begin{matrix} \\ \\ \\ \\ \\ Q \end{matrix} \begin{matrix} \\ \\ \\ \\ \\ R \end{matrix}$$

$\dim_{\mathbb{R}}(V_1 + V_2) = \dim_{\mathbb{R}} V_1 + \dim_{\mathbb{R}} V_2 - \dim_{\mathbb{R}}(V_1 \cap V_2)$

$R = P + 3Q \in V_1$

$T = 2P + 3Q + S \in V_2$

$U = P + 4Q + S$

$R - P - 3Q = 0$

$T - S = 2P + 3Q \Rightarrow T - S \in V_1 \cap V_2$

$U - S = P + 4Q \Rightarrow U - S \in V_1 \cap V_2$

$T - S \in V_2$

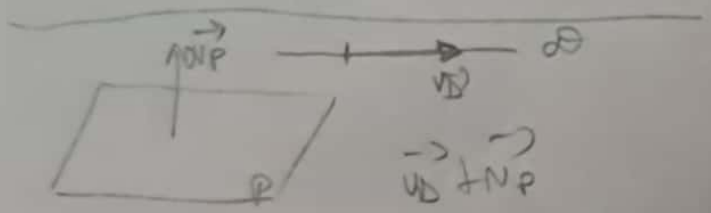
$2P + 3Q \in V_1$

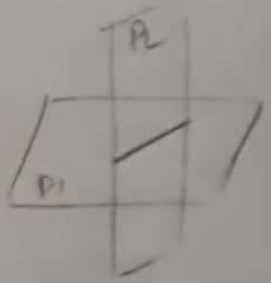
$U - S \in V_2$

$P + 4Q \in V_1$

$\Rightarrow$ baza pt $V_1 \cap V_2 = \{T - S, U - S\}$

$$\begin{cases} x_1 - 10x_3 - x_4 - 5x_2 = 0 \\ x_5 + 13x_3 + 7x_2 = 0 \end{cases} \Leftrightarrow \begin{cases} x_1 = 10 \cdot \dots \\ x_5 = \dots \end{cases}$$





~~Exercitiu~~ 1 rezolvat

① $f: \mathbb{R}_2[X] \to \mathcal{M}_2(\mathbb{R})$

$$M_{B_1,B_2}(f) = A = \begin{pmatrix} -1 & 2 & 3 \\ 1 & 0 & -1 \\ 1 & 4 & 3 \\ -5 & 4 & 9 \end{pmatrix} \in \mathcal{M}_{4,3}(\mathbb{R})$$

$B_1 = \{1, 1+X, 1+X^2\}$

$B_2 = \left\{E_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, E_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, E_3 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, E_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right\}$

a) Fie $b_1' = \{1, X, X^2\}$, baza canonica pt $\mathbb{R}_2[X]$, resp $\mathcal{M}_2(\mathbb{R})$.

$B_2' = B_2$

$M_{B_1',B_2'}(f) = C^{-1} \cdot A \cdot M_{B_1 \to B_1'}$; $C \overset{not}{=} M_{B_2 \to B_2'}$

$$C^{-1} = M_{B_2' \to B_2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$M_{B_1 \to B_1'} = M_{B_1' \to B_1}^{-1} = \underbrace{\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\Delta}^{-1} \overset{not}{=} \Delta^{-1}$$

Let $\Delta = 1$

$\Rightarrow \Delta^{-1} = \Delta^t$

$${}^t\Delta = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \Rightarrow \Delta^t = \begin{pmatrix} 1 & -1 & -1 \\ 0 & +1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \Delta^{-1} = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow M_{B_1',B_2'}(f) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 2 & 3 \\ 1 & 0 & -1 \\ 1 & 4 & 3 \\ -5 & 4 & 9 \end{pmatrix} \begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 2 & 3 \\ 1 & 0 & -1 \\ 1 & 4 & 3 \\ -5 & 4 & 9 \end{pmatrix} \begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow M_{B_1',B_2'}(f) = \begin{pmatrix} -1 & 3 & 4 \\ 1 & -1 & -2 \\ 1 & 3 & 2 \\ -5 & 9 & 14 \end{pmatrix} \begin{matrix} f(1) \\ f(X) \\ f(X^2) \end{matrix}$$

$$\Rightarrow f(a_0 + a_1X + a_2X^2) = a_0\begin{pmatrix} -1 & 1 \\ 1 & -5 \end{pmatrix} + a_1\begin{pmatrix} 3 & -1 \\ 3 & 9 \end{pmatrix} + a_2\begin{pmatrix} 4 & -2 \\ 2 & 14 \end{pmatrix}$$

b) $\operatorname{Im} f = \mathcal{L}\left\{\begin{pmatrix} -1 & 1 \\ 1 & -5 \end{pmatrix}, \begin{pmatrix} 3 & -1 \\ 3 & 9 \end{pmatrix}, \begin{pmatrix} 4 & -2 \\ 2 & 14 \end{pmatrix}\right\}$

P Q R

$$\begin{pmatrix} -1 & 3 & 4 \\ 1 & -1 & -2 \\ 1 & 1 & 2 \\ -5 & 9 & 14 \end{pmatrix} \xrightarrow[L_4 \to L_4 - 5L_1]{\substack{L_2 \to L_2 + L_1 \\ L_3 \to L_3 + L_1}} \begin{pmatrix} -1 & 3 & 4 \\ 0 & 2 & 2 \\ 0 & 6 & 6 \\ 0 & -6 & -6 \end{pmatrix} \xrightarrow[L_4 \to L_4 \left(-\frac{1}{6}\right)]{\substack{L_2 \to L_2 \frac{1}{2} \\ L_3 \to L_3 \frac{1}{6}}}$$

$$\to \begin{pmatrix} -1 & 3 & 4 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow[\substack{L_4 \to L_4 - L_2 \\ L_1 \to L_1 + (-1)}]{L_3 \to L_3 - L_2} \begin{pmatrix} +1 & -3 & -4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{L_1 \to L_1 + 3L_2} \begin{matrix} P & Q & R \\ \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{matrix}$$

$\{P, Q, R\}$ SG

$\Rightarrow \{P, Q, R\}$ SLD , $\{P, Q\}$ SLi $\Big]$ $\Rightarrow$ o bază pt $\operatorname{Im} f$ poate fi $\{P, Q\}$

$\Rightarrow \dim_{\mathbb{R}} \operatorname{Im} f = 2 \to \dim_{\mathbb{R}} \operatorname{Ker} f = 1$

$\operatorname{Ker} f = \{ a_0 + a_1 X + a_2 X^2 \mid f(a_0 + a_1 X + a_2 X^2) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \}$ ★

$\Rightarrow \operatorname{Ker} f = \{ a_0 P + a_1 Q + 2a_2 R \mid f(\quad) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \}$

$\overset{\text{Egalăm}}{\underset{\star}{=}}$ $\begin{matrix} P = R \\ Q = -R \end{matrix}$

$\Rightarrow \operatorname{Ker} f = \{ a_0 \cdot R + (-1) a_1 \cdot R + 2a_2 R \mid a_0, a_1, a_2 \in \mathbb{R} \}$

$= \{ \underbrace{(a_0 - a_1 + 2a_2)}_{\text{notăm } \alpha \in \mathbb{R}} R \mid a_0, a_1, a_2 \in \mathbb{R} \}$

$\Rightarrow \operatorname{Ker} f = \operatorname{sp}\{R\}$

$\{R\}$ SG

$\dim_{\mathbb{R}} \{R\} = 1$

$\dim_{\mathbb{R}} \operatorname{Ker} f = 1$

$\Big]$ $\Rightarrow$ o bază pt $\operatorname{Ker} f$ poate fi $\{R\}$

c) $V \subseteq \mathcal{M}_2(\mathbb{R})$; $\dim_{\mathbb{R}} V = 2$; $V \cap \operatorname{Im} f = \operatorname{sp}\{\begin{pmatrix} 3 & -1 \\ 3 & 0 \end{pmatrix}\}$

$\dim_{\mathbb{R}} \operatorname{Im} f = 2$; $\dim_{\mathbb{R}} (V \cap \operatorname{Im} f) = 1$

$\overset{\text{Grassmann}}{\Rightarrow} \dim_{\mathbb{R}} (V + \operatorname{Im} f) = 3$.

$\operatorname{Im} f = \operatorname{sp}\{\begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 3 & -1 \\ 3 & 0 \end{pmatrix}\}$ (conf. b)

$V = ?$, $\dim_{\mathbb{R}} V = 2$

$V \cap \operatorname{Im} f = \operatorname{sp}\{\begin{pmatrix} 3 & -1 \\ 3 & 0 \end{pmatrix}\}$

$\to$ fie $\begin{pmatrix} 0 & \frac{1}{2} \\ 0 & 2 \end{pmatrix} \in \mathcal{M}_2(\mathbb{R}) \setminus V$

$\Rightarrow V = \operatorname{sp}\{\begin{pmatrix} 3 & -1 \\ 3 & 0 \end{pmatrix}, ?\}$

$V + \operatorname{Im} f$

$$\begin{pmatrix} -1 & 3 & 3 & a \\ 1 & -1 & -1 & b \\ 1 & 3 & 3 & c \\ -5 & 9 & 9 & d \end{pmatrix} \xrightarrow[\substack{L_3 \to L_3 + L_2 \\ L_4 \to L_4 - 5L_2}]{L_2 \to L_2 + L_1} \begin{pmatrix} -1 & 3 & 3 & a \\ 0 & 2 & 2 & a+b \\ 0 & 6 & 6 & c+a \\ 0 & -6 & -6 & d-5a \end{pmatrix} \xrightarrow[\substack{L_3 \to \frac{1}{6} L_3 \\ L_4 \to -\frac{1}{6} L_4}]{L_2 \to \frac{1}{2} L_2}$$

$$\begin{pmatrix} 0 & 1 & 1 & \frac{a}{2} \\ 0 & 1 & 1 & \frac{a+b}{2} \\ 0 & 1 & 1 & \frac{c+a}{6} \\ 0 & 1 & 1 & \frac{5a-d}{6} \end{pmatrix} \xrightarrow[L_4 \to L_4 - L_2]{L_3 \to L_3 - L_2} \begin{pmatrix} -1 & 3 & 3 & a \\ 0 & 1 & 1 & \frac{a+b}{2} \\ 0 & 0 & 0 & \frac{c+a}{6} - \frac{a+b}{2} \\ 0 & 0 & 0 & \frac{5a-d}{6} - \frac{a+b}{2} \end{pmatrix}$$

$$\xrightarrow{L_1 \to -L_1} \begin{pmatrix} +1 & -3 & -3 & -a \\ 0 & 1 & 1 & \frac{a+b}{2} \\ 0 & 0 & 0 & \frac{c-7a+3b}{6} \\ 0 & 0 & 0 & \frac{2a-d+3b}{6} \end{pmatrix} \xrightarrow{L_1 \to L_1 + 3L_2} \begin{pmatrix} 1 & 0 & 0 & \frac{3a+3b}{2} - a = \frac{a+3b}{2} \\ 0 & 1 & 1 & \frac{a+b}{2} \\ 0 & 0 & 0 & \frac{c-2a+3b}{6} \\ 0 & 0 & 0 & \frac{2a-d+3b}{6} \end{pmatrix}$$

$\dim_{\mathbb{R}}(V + \operatorname{Im} f) = 3 \Rightarrow$ $\frac{c-2a+3b}{6} = 1$ (compatibil)

$\frac{2a-d+3b}{6} = 0$; $\frac{a+b}{2} = 0$ (compatibil), $\frac{a+3b}{2} = 0$ (compatibil)

$\Rightarrow a+b=0$ $\Rightarrow a+3b=0$

$\Rightarrow \begin{cases} c-2a+3b=6 \\ 2a-d+3b=0 \Rightarrow d = 2a+3b \Rightarrow d=b \end{cases}$ ④ ⑤

$\Rightarrow c-d+6b=6$ $\quad c-4a+d=6$ ⑥

$\Rightarrow c-d+6b-c+4a-d=0$

$4a+6b-2d=0$

b

$4a+4b=0$

$\begin{cases} a+b=0 \Rightarrow b=-a \\ a+b=0 \end{cases} \Big| \cdot (-2) \Rightarrow -2a=0 \Rightarrow a=0$

$\Rightarrow b=0$

$\Rightarrow d=0$

$\Rightarrow \boxed{c=6}$

$\Rightarrow$ Vpropte f: $\operatorname{sp}\left\{ \begin{pmatrix} 3 & -1 \\ 3 & 3 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 6 & 0 \end{pmatrix} \right\}$??

⑤ $f: \mathbb{R}^3 \to \mathbb{R}^3$; $f(1,0,1) = (1,1,2) \Rightarrow f(e_1) + 2f(e_3) = (1,1,2)$

$f(0,1,1) = (-1,0,1) \Rightarrow f(e_2) + 2f(e_3) = (-1,0,1)$

$\ker f = \{(1,1,1)\} = \{\alpha(1,-1,1) \mid \alpha \in \mathbb{R}\} = \{(\alpha, -\alpha, \alpha) \mid \alpha \in \mathbb{R}\}$.

f izomorfism $\Leftrightarrow M_{\mathcal{B}_c, \mathcal{B}_c}(f)$ inv.

$\det = 0 \Rightarrow f(1,-1,1) = (0,0,0) \Rightarrow f(e_1) - f(e_2) + f(e_3) = (0,0,0)$

$\Rightarrow (1,-1,1) \in \ker f$

$\Rightarrow f(e_1) - f(e_2) = (2,1,1) \mid \cdot 2 \Rightarrow 2f(e_1) - 2f(e_2) = (4,2,2)$

$\begin{cases} f(e_1) + 2f(e_3) = (1,1,2) \\ f(e_2) + 2f(e_3) = (-1,0,1) \end{cases} \Rightarrow -2f(e_1) + 3f(e_2) = (-1,0,1)$

$\begin{cases} f(e_1) + 2f(e_3) = (1,1,2) \end{cases}$

$\Rightarrow \boxed{f(e_2) = (3,2,3)}$

$\Rightarrow \boxed{f(e_1) = (5,3,4)}$

$\Rightarrow f(e_3) = (1,1,2) - (3,2,3) = (-4,-2,-2)$

$\Rightarrow \operatorname{Im} f = \mathcal{L}\{(5,3,4),(3,2,3),(-2,-1,1)\}$

$$M_{BC}(f) = \begin{pmatrix} 5 & 3 & -2 \\ 3 & 2 & -1 \\ 4 & 3 & -1 \end{pmatrix} \xrightarrow{L_1 \to L_1 - L_2} \begin{pmatrix} 2 & 1 & -1 \\ 3 & 2 & -1 \\ 4 & 3 & -1 \end{pmatrix} \xrightarrow[L_3 \to L_3 - L_4]{L_2 \to L_2 - 3L_4} \begin{pmatrix} 2 & 1 & -1 \\ -5 & -4 & 2 \\ 4 & 3 & -1 \end{pmatrix}$$

$$\to \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 2 \\ 0 & 3 & 3 \end{pmatrix} \xrightarrow[L_3 \to L_3 - \frac{3}{2}L_2]{L_2 \to \frac{1}{2}L_2} \begin{pmatrix} -1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow[\cdot(-1)]{L_3 \to L_3 - L_2} \begin{pmatrix} -1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$\Rightarrow$ baza pt $\operatorname{Im} f$ poate fi $\{(5,3,4),(3,2,3)\}$

$$\det(M_{BC}(f)) = \begin{vmatrix} 5 & 3 & -2 \\ 3 & 2 & -1 \\ 4 & 3 & -1 \end{vmatrix} = -10 - 12 - 12 + 16 + 15 + 9$$

$= 0 \Rightarrow f$ nu este izomorfism ??

brut

Verif: ①

$$V + \operatorname{Im} f \quad \begin{pmatrix} 3 & 0 & -1 & 3 \\ -1 & 0 & -1 & -1 \\ 3 & 6 & -1 & 3 \\ 9 & 0 & -5 & 9 \end{pmatrix} \to \begin{pmatrix} -1 & 0 & 3 & 3 \\ -1 & 0 & -1 & -1 \\ -1 & 6 & 3 & 3 \\ -5 & 0 & 9 & 9 \end{pmatrix} \xrightarrow[L_3 \to L_3 + L_1]{L_2 \to L_2 + L_1} \begin{pmatrix} -1 & 0 & 3 & 3 \\ -2 & 0 & 2 & 2 \\ 0 & 6 & 6 & 6 \\ -4 & 0 & 12 & 12 \end{pmatrix}$$

$$\to \begin{pmatrix} -1 & 0 & 3 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 6 & 6 & 6 \\ 0 & 0 & -6 & -6 \end{pmatrix} \to \begin{pmatrix} -1 & 0 & 3 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & -6 & -3 & -3 \\ 0 & 0 & -6 & -6 \end{pmatrix}$$

$$\to \begin{pmatrix} -1 & 3 & 0 & 3 \\ 0 & -6 & -3 & -3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -6 & -6 \end{pmatrix} \xrightarrow[L_4 \to L_4 - L_2]{L_3 \to L_3 - L_2} \begin{pmatrix} -1 & 3 & 0 & 3 \\ 0 & -6 & -3 & -3 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$u_1 \; v_1 \; v_2 \; u_2$ ✓

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⑥ $f, h : \mathbb{R}^2 \to \mathbb{R}^3$

$\Rightarrow g : \mathbb{R}^2 \to \mathbb{R}^2$ $f : \mathbb{R}^2 \to \mathbb{R}^3$

$f \circ g = h$; $f(g) = h$ $\Rightarrow f(g(x,y,z)) = h(x,y,z)$

$f(x,y) = (x-y, 2x+y, 3x+y)$ $\Rightarrow$

$h(x,y) = (x+2y, 2x+3y, 3x+4y)$ $f(x,y) = x(1,2,3) + y(-1,1,1)$

$f \circ g = h \Rightarrow g : \mathbb{R}^3 \to \mathbb{R}^2$; $g(x,y,z) = ?$

$f(g(x,y,z)) = h(x,y,z) \mid \circ f^{-1}$??

$\Rightarrow g(x,y,z) = f^{-1} \circ h = f^{-1}(h(x,y,z))$

$$M_{BC}(f) = \begin{pmatrix} 1 & 2 & 3 \\ -1 & 2 & 1 \end{pmatrix} \to$$

$g: \mathbb{R}^2 \to \mathbb{R}^2$, $g(x,y,z) = (x-y+z, z, y)$

$g: \mathbb{R}^2 \to \mathbb{R}^2$, $g(x,y,z) = (x-y+z, y-z)$

$g \circ f = g(f(x,y,z)) = g(x-y+z, z, y) = (x-y+z+z+y, z-y)$

$\Rightarrow (g \circ f)(x,y,z) = (x+y+2z, z-y)$ $\underline{\neq}$ not $f(x,y,z): \mathbb{R}^3 \to \mathbb{R}^2$

$B_1 = \{(1,0,0), (1,1,0), (1,1,1)\}$ G. $\mathbb{R}^3$

$B_2 = \{(1,1), (1,0)\}$ G. $\mathbb{R}^2$

Fie $B_1' = \{\underset{e_1}{(1,0,0)}, \underset{e_2}{(0,1,0)}, \underset{e_3}{(0,0,1)}\}$ G. $\mathbb{R}^3 = (B_C)_{\mathbb{R}^3}$

$B_2' = \{\underset{e_4}{(1,0)}, \underset{e_5}{(0,1)}\}$ G. $\mathbb{R}^2 = (B_C)_{\mathbb{R}^2}$

$$M_{B_1, B_2'}^{(f)} = \begin{pmatrix} f(e_1) & f(e_2) & f(e_3) \\ 1 & 1 & 2 \\ 0 & -1 & 1 \end{pmatrix}$$

$f(x,y,z) = x(1,0) + y(1,-1) + z(2,1)$

$$M_{B_1, B_2'}^{(f)} = M_{B_2 \to B_2'} \cdot M_{B_1, B_2}^{(f)} \cdot M_{B_1 \to B_1'}$$

$M_{B_1 \to B_1'} = (M_{B_1' \to B_1})^{-1}$

$M_{B_2' \to B_2} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$

$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}^{-1}$

$\Rightarrow \begin{pmatrix} 1 & 1 & 2 \\ 0 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \cdot M_{B_1, B_2}^{(f)} \cdot \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}^{-1} / \cdot \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^{-1}$ M

$\cdot \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ dr

$\Rightarrow M_{B_1, B_2}^{(f)} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 1 & 2 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

$\begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$

$^t\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$

$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^t = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$

$$M_{B_1, B_2}^{(f)} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 3 & 4 \end{pmatrix}$$ ✓

⑧ $f: \mathbb{R}^3 \to \mathbb{R}^3$; $f = ?$

$f(1,0,0), f(1,1,0), f(1,1,1)$

$$M_{B_C}(f) = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$\Rightarrow f(x,y,z) = x(1,1,0) + y(1,0,1) + z(0,1,1)$

Fie $(B_C)_{\mathbb{R}^3} \overset{not}{=} B_1 = \{\underset{e_1}{(1,0,0)}, \underset{e_2}{(0,1,0)}, \underset{e_3}{(0,0,1)}\}$

$B = \{(1,1,1), (2,1,1), (1,1,2)\} \Rightarrow$

$f(1,2,1) = f(e_1) + 2f(e_2) + f(e_3) = (1,1,0) + 2(1,0,1) + (0,1,1)$

$= (3,2,3)$

$M_{B_1,B_1'}(f) = M_{B\to B_1'} \cdot M_B(f) \cdot M_{B\to B_1}$

$B_1' = B_1 - 5e$
$B_2' = B_1$
$B_1 = B$
$B_2 = B$

$\Rightarrow M_{B_1,B_2'}(f) = M_{B_2\to B_2'}^{-1} \cdot M_{B_1,B_2'}(f) \cdot M_{B_1\to B_1'}$

$\Rightarrow M_{B_1}(f) = M_{B\to B_1}^{-1} \cdot M_B(f) \cdot M_{B_1\to B}^{-1}$

$M_{B_1\to B} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 3 & 1 \end{pmatrix}$

$\Rightarrow M_{B_1}(f) = M_{B_1\to B}^{-1} \cdot M_B(f)$

$\Rightarrow M_B(f) = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 3 & 1 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 3 & 1 \end{pmatrix}$

$\det \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 3 & 1 \end{pmatrix} = 1 + 6 + 2 - 1 - 3 - 4 = 1$

$t(\ ) = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 1 & 1 & 1 \end{pmatrix}$

$^*(\ ) = \begin{pmatrix} -2 & 1 & 1 \\ -1 & 0 & 1 \\ 5 & -1 & -3 \end{pmatrix}$

$\Rightarrow M_B(f) = \begin{pmatrix} -2 & 1 & 1 \\ -1 & 0 & 1 \\ 5 & -1 & -3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 3 & 1 \end{pmatrix}$

$= \begin{pmatrix} -1 & 3 & 0 \\ 0 & 1 & 0 \\ 4 & -2 & 2 \end{pmatrix}$ //

$B = \{(1,2,1), (2,1,3), (1,1,1)\}$
$e_4 \quad e_5 \quad e_6$

$M_B(f) = \begin{pmatrix} f(e_4) & f(e_5) & f(e_6) \\ & & \\ & & \end{pmatrix}$

$B_1 = \{(1,2,1), (2,1,1), (1,0,1)\}$

$M_{B_1}(f) = \begin{pmatrix} f(e_1) & f(e_2) & f(e_3) \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$

$f(e_1) = (1,1,0)$
$f(e_2) = (1,0,1)$
$f(e_3) = (0,1,1)$

$[f(e_1)]_B = M_{B,B_1}(f) \cdot [e_1]_{B_1}$

$= \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 3 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$

$\begin{cases} \alpha + 2\beta + \gamma = 1 \\ 2\alpha + \beta + \gamma = 0 \\ \alpha + 3\beta + \gamma = 0 \end{cases}$

$\begin{cases} -3\beta - \gamma = 2 \\ -\beta - \gamma = 0 \end{cases}$

$f: V_3 \to V_3,\ f(\vec{x}) = (\vec{x} + \vec{d}) \times \vec{x}$

f op lin

$\Rightarrow f(\alpha\vec{x} + \beta\vec{y}) = \alpha f(\vec{x}) + \beta f(\vec{y}) \quad \forall \alpha, \beta \in \mathbb{R}\ \forall \vec{x}, \vec{y} \in V_3$

$f(\alpha\vec{x} + \beta\vec{y}) = (\vec{x} + \vec{0} - \vec{k}) \times (\alpha\vec{x} + \beta\vec{y})$

$= -(\alpha\vec{x} + \beta\vec{y}) \times (\vec{x} + \vec{0} - \vec{k})$

$= -\left[(\alpha\vec{x}) \times \vec{x} + (\beta\vec{y}) \times \vec{z}\right]$

$= -\left[\alpha(\vec{x} \times \vec{z}) + \beta(\vec{y} \times \vec{z})\right]$

$= -\left[-\alpha(\vec{z} \times \vec{x}) - \beta(\vec{z} \times \vec{y})\right]$

$= \alpha(\vec{z} \times \vec{x}) + \beta(\vec{z} \times \vec{y})$

$= \alpha f(\vec{x}) + \beta f(\vec{y})$

$\Rightarrow f$ op lin.

$f(\vec{i}) = (\vec{i} + \vec{d} - \vec{k}) \times \vec{i}$

$= \begin{vmatrix} \vec{i} & \vec{d} & -\vec{k} \\ 1 & 0 & 0 \end{vmatrix} = \vec{i} \cdot 0 + \vec{d} \cdot (-1) \cdot 1 + \vec{k} \cdot (-1)$

$= -\vec{d} - \vec{k} = -(\vec{d} + \vec{k})$

$f(\vec{d}) = (\vec{i} + \vec{d} - \vec{k}) \times \vec{d}$

$= \begin{vmatrix} \vec{i} & \vec{d} & -\vec{k} \\ 0 & 1 & 0 \end{vmatrix} = \vec{i} \cdot 1 + \vec{d} \cdot 0 + \vec{k} \cdot 1$

$= \vec{i} + \vec{k}$

$f(\vec{k}) = \begin{vmatrix} \vec{i} & \vec{d} & -\vec{k} \\ 0 & 0 & 1 \end{vmatrix} = \vec{i} \cdot 1 + \vec{d} \cdot (-1) + \vec{k} \cdot 0$

$= \vec{i} - \vec{d}$

$\Rightarrow M_{(\vec{i},\vec{d},\vec{k})}(f) = \begin{pmatrix} -1 & 1 & 1 \\ -1 & 0 & -1 \\ 0 & -1 & -1 \end{pmatrix}$

$\operatorname{Im} f = \langle f(\vec{i}), f(\vec{d}), f(\vec{k}) \rangle$

$\begin{pmatrix} 0 & 1 & 1 \\ -1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix} \longrightarrow \begin{pmatrix} -1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{L_2 \to L_2 - L_1} \begin{pmatrix} -1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{L_3 \to L_3 - L_2}$

$\begin{pmatrix} -1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ base $\operatorname{Im} f = \{f(\vec{i}), f(\vec{d})\}$, $\dim \operatorname{Im} f = 2$

$\dim \ker f = 0$

$\ker f = \{ \vec{a} \mid f(\vec{a}) = \vec{0} \}$

$f(\vec{a}) = (\vec{i} + \vec{j} - \vec{k}) \times \vec{a} = \vec{0} \Rightarrow \vec{a} = \vec{0}$

$\Rightarrow \ker f = \{\vec{0}\} \Rightarrow \dim_{\mathbb{R}} \ker f = 0$

$\Rightarrow f$ injectivă.

f surj $\Leftrightarrow \operatorname{Im} f = V_3$

$\left.\begin{array}{l} \operatorname{Im} f \subset V_3 \\ \dim_{\mathbb{R}} V_3 = 3 \\ \dim_{\mathbb{R}} \operatorname{Im} f = 3 \end{array}\right| \Rightarrow \operatorname{Im} f = V_3 \Rightarrow f$ surj